Brain-age error analysis — best LaBraM fine-tune vs. the data

📄 **PDF version of this notebook** (figures, tables and formulas, no code).

Where, and why, does the brain-age model make its errors? This notebook takes the best short-schedule run (selected on **validation** case MAE) and examines its errors against the data:

1. **Training dynamics**: the train/val gap (memorization).
2. **Error summary**: window vs. recording level, mean vs. median, by cohort (normal / abnormal), including the train split scored in eval mode.
3. **Error and loss distributions**: per sample (window) and per recording, and the tails.
4. **Per-sample view**: spread of the window predictions inside a recording, mean vs. median pooling, and position in the recording.
5. **Regression to the mean**: error by age bin against the train sampling density (how over- or under-sampled each age is).
6. **Post-hoc bias correction**: fitted on val, applied to test.
7. **Normal vs. abnormal**, and the **spectral correlates** of the residual.
8. **Real EEG**: at the same true age, a recording predicted accurately vs. one predicted too young vs. one too old, side by side; the extreme cases; an **interactive explorer**; and a playground.
9. **Run comparison**: predicted vs. true age per decade across the other trained runs (mixup / C-Mixup / EMA / long / age-balanced loss).
10. **TUEG**: the same TUAB test recordings scored by models trained on TUAB + TUEG (the whole TUH EEG corpus). Does more, and more varied, training data change the error pattern?

Which model is analysed, and why — the baselines

The brain-age study (`docs/age_overfitting_experiments.md`) grew in steps. Each run below is the previous one with one change; all use the same subject-disjoint TUAB split (2,145 train / 556 val / 274 test recordings). The numbers are recording MAE in years (Eq. (1.5)), val / test:

- **A**: LaBraM's own fine-tuning recipe adapted to age regression, scored on whole recordings. 9.03 test.
- **B**: A, scored on the first 5 minutes of each recording after 60 s trims, so evaluation is cheaper and less noisy. 8.40 / 8.91.
- **D**: B plus a **common average reference** (CAR) on the input. This is the default `finetune_tuab_age.json` , "scenario D": Huber δ = 1, lr 1e-4, layer decay 0.65, 15 epochs. 7.77 / 8.30. CAR was the one large gain of the early ablations.

- **E2**: D on a **4-epoch** schedule with 1 warmup epoch. D's best epoch was already epoch 1–3, after which it memorized the training set. 7.68 / 8.25.
- **M1**: E2 plus **mixup** α = 0.4, the default analysed here. 7.46 / 8.45.

Is D the best choice? No. **D is the reference configuration**, the baseline every TUAB ablation was compared against; it is not the best model.

- **Within TUAB-only training:** the run to analyse is chosen on **validation** MAE, because the test set must not choose the model. M1 has the lowest val MAE of all TUAB-only runs (7.46; D 7.77). A few runs score slightly lower on test (M2 8.17, E1c 8.13), but every difference among E2, M1, M2 and E1c is within run-to-run noise (about ±0.4 years). The error pattern this notebook studies is the same for all of them (Section 9).
- **Overall:** the best models are **not** TUAB-only. Training on TUAB + TUEG and cross-validating lowers the TUAB test MAE to 7.10–7.44 (`notebooks/brain_age_research_summary.ipynb`). Section 10 compares them with M1 recording by recording.

Setup of M1: scenario D (CAR input, Huber δ=1, lr 1e-4, layer decay 0.65) on the 4-epoch schedule with mixup, subject-disjoint split, 5-minute val/test budget per recording (30 × 10 s windows), 60 s trimmed at each end. A recording's prediction is the mean of its window predictions, which is the metric reported in ClearML. Predictions are cached in `<run_dir>/analysis/`.

Conventions

Every loss, metric and derived value is defined by a numbered equation, written as the code computes it, before it is first used. Equations, tables, figures and figure panels are numbered per section: Eq. (2.1), Table 2.2, Figure 3.1(b). Figures have at most two panels side by side. Symbols used throughout:

symbol	meaning
r, w	a recording; a 10 s window of a recording
y_r	true age of recording r (years, from the EDF header); a window inherits $y_{r,w} = y_r$
$\hat{y}_{r,w}$	the model's prediction for window w (years)
$\hat{y}_r$	the recording prediction, the mean of its window predictions, Eq. (0.10)
$e = \hat{y} - y$	signed error (years); $e > 0$ means predicted too old
μ_y, σ_y	mean and sample std of the train-window ages (target z-scoring, Eq. (0.2))
S, n	the set being scored (windows or recordings of one split/cohort) and its size

How to use it

- **Open**: in VS Code (Remote-SSH to this box), open `notebooks/age_error_analysis.ipynb` and pick the kernel **"LaBraM (.venv)"**. Or run JupyterLab on the box, `cd /data/code/LaBraM && .venv/bin/jupyter lab --no-browser --port 8888`, then tunnel it: `ssh -L 8888:localhost:8888 <box>` and open the printed `http://localhost:8888/...` link.
- **Run**: *Run All* takes about 2 minutes. Predictions are cached per run, and the GPU is needed only the first time a run is scored.

- **Play**: change the **Parameters** cell (which run, which ages to compare, which runs to overlay) and re-run. Every section after **0. Load** is independent, so re-run just the one you are editing. Section 8 has an interactive EEG browser and a playground cell for your own queries.
- **Code**: analysis helpers are in `labram/eval/age_analysis.py`; EEG plots and the explorer are in `labram/eval/age_plots.py`. Cells call them with plain arguments, so changing an argument in a cell is usually all you need.

Parameters

Setup

0. Load the run and its predictions

The model is rebuilt from the run's own `run_config.yaml`, with the same window selection, preprocessing and EMA choice it was evaluated with. Every window keeps its dataset index, so any prediction can be traced back to its EEG. Table 0.1 counts the scored windows and recordings per split.

```
run           : age_M1_short_mixup04_20261007_231716_154
weights       : raw | checkpoint-best.pth
schedule      : 4 epochs, 1 warmup, lr 0.0001 , layer decay 0.65
mixup         : True alpha 0.4 sigma 0.0
loss          : huber delta 1.0 | balance none
target stats  : mean 48.88, std 17.82 years (train windows)
channels      : FP1, FP2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, A1, A
2, FZ, CZ, PZ, T1, T2
```

Table 0.1. Scored windows and recordings per split

The train split is scored on `TRAIN_WINDOWS_PER_RECORDING` windows per recording.

split	windows	recordings
test	8220	274
train	21450	2145
val	16680	556

0.1 Experiment configuration and training objective

This subsection states everything the run in `BEST_RUN` was trained with (default: `age_M1_short_mixup04_20261007_231716_154`): data, input, model, target, loss, mixup, optimizer, schedule and model selection. Table 0.2 (below) is generated from the run's own `run_config.yaml`; each row cites the equation it enters. Config fields not listed (`loss.amplitude_weight`, `loss.phase_weight`, `loss.vq_commitment_beta`,

`loss.classification_label_smoothing`, `loss.soft_label_*`, `model.codebook_reg.*`, `model.lora.*`, `optimizer.smoothing`) belong to other tasks or to switched-off features and have no effect on this run.

Data. TUAB v3.0.0 EDFs, band-passed 0.1–75 Hz, 50 Hz notch, resampled to 200 Hz, $C = 23$ channels (`dataset_maker/make_TUAB.py`), cut into 10 s windows $x_{r,w} \in \mathrm{R}^{C \times 2000}$ (μV). `trim_start_sec = trim_end_sec = 60` drops the first and last minute of every recording. Train uses every remaining window; val/test use the first `eval_minutes = 5` after the trim, $W = 30$ windows. The split is subject-disjoint (`data/age_splits.py`).

Input. The loader divides by 100 (units of 100 μV) and the model applies a common average reference (`labram_plus.common_average_reference`; per-patch z-scoring is off):

$$\tilde{x}_{c,t} = \frac{x_{c,t}}{100} - \frac{1}{C}\sum_{c'=1}^{C}\frac{x_{c',t}}{100}$$

The window is cut into $A = 10$ patches of 200 samples (1 s) per channel, embedded by LaBraM-Base (`labram_base_patch200_200`: 12 transformer blocks, width 200, 10 heads, absolute position and time embeddings, no relative position bias, no qkv bias), mean-pooled over the $C \cdot A$ patch tokens, layer-normed (`fc_norm`) and mapped to one scalar by a linear head $\hat{z} = f_\theta(\tilde{x})$. The head's initial weights are scaled by `init_scale` = 0.001, so training starts from $\hat{z} \approx 0$, i.e. predicting μ_y.

Target. Ages are z-scored with the train-window statistics; predictions go back to years before any metric is computed:

$$z = \frac{y - \mu_y}{\sigma_y}, \qquad \mu_y = \frac{1}{N}\sum_{i=1}^{N} y_i, \quad \sigma_y = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \mu_y)^2}$$

$$\hat{y} = \mu_y + \sigma_y \hat{z}$$

where i runs over the N selected train windows.

Mixup (`mixup.enabled`, `alpha` = 0.4, `prob` = 1, `sigma` = 0). Every training batch of $B = 64$ windows is blended with a random permutation π of itself, inputs and targets with the same λ, drawn once per batch:

$$\lambda \sim \mathrm{Beta}(\alpha, \alpha), \qquad \tilde{x}_i' = \lambda \tilde{x}_i + (1-\lambda)\tilde{x}_{\pi(i)}, \qquad z_i' = \lambda z_i + (1-\lambda) z_{\pi(i)}$$

With $\alpha = 0.4$ most draws sit near 0 or 1 (light mixing), some near 0.5. `sigma` = 0 means plain mixup; C-Mixup ($\sigma > 0$, run M1c) instead draws the partner j of i with probability $\propto \exp\!\left(-(y_i - y_j)^2 / 2\sigma^2\right)$.

Loss (`loss.regression_loss = huber`, `huber_delta` $\delta = 1$, no age balancing). The Huber loss of the z-scored residual $u = \hat{z}_i - z_i'$:

$$\ell_\delta(u) = \begin{cases} \frac{1}{2}u^2 & |u| \le \delta \\ \delta\left(|u| - \frac{1}{2}\delta\right) & |u| > \delta \end{cases}$$

$$\mathrm{L} = \frac{1}{B}\sum_{i=1}^{B} \ell_\delta\left(\hat{z}_i - z_i^{'}\right)$$

In years the loss is quadratic for errors up to $\delta\sigma_y$ (about 18 y) and linear beyond, so a few very wrong windows cannot dominate the gradient.

Optimizer (`opt = adamw` , $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, `weight_decay` $\lambda_{wd} = 0.05$). For a parameter θ in group j at step t with gradient g_t:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)\, g_t, \qquad v_t = \beta_2 v_{t-1} + (1-\beta_2)\, g_t^2, \qquad \hat{m}_t = \frac{m_t}{1-\beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1-\beta_2^t}$$

$$\theta_t = \theta_{t-1} - \eta_{t,j}\left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda_{wd,j}\,\theta_{t-1}\right)$$

$\lambda_{wd,j} = 0$ for biases, norm weights (every 1-D tensor) and `pos_embed` , `cls_token` , `time_embed` ; 0.05 otherwise. Gradients are not clipped (`clip_grad` = null); training runs in fp16 autocast with dynamic loss scaling.

Layer-wise learning-rate decay (`layer_decay` $\gamma = 0.65$). Group $j = 0$ is the patch embedding, `pos_embed` and `cls_token` ; $j = 1 \ldots 12$ the transformer blocks; $j = 13$ `fc_norm` , the head and every other parameter, which includes `time_embed` (`optim_factory.get_num_layer_for_vit` has no rule for it, so the temporal embedding trains at the full rate):

$$\eta_{t,j} = \eta_t \gamma^{13-j}$$

so the head learns at the full rate, the last block at $0.65\times$, the first block at $0.65^{12} \approx 0.006\times$ and the embeddings at $0.65^{13} \approx 0.004\times$: the pre-trained representation is changed mostly near the top.

Schedule (`sched = cosine` , `lr` $\eta_{\max} = 10^{-4}$, `min_lr` $\eta_{\min} = 10^{-6}$, `warmup_epochs` = 1, `epochs` = 4; T iterations in total, T_w of them warmup, both counted in optimizer steps). The warmup starts at 0: `optimizer.warmup_lr` is not passed to the schedule (`runs/common.py::make_lr_schedule`), so it has no effect:

$$\eta_t = \begin{cases} \eta_{\max} \dfrac{t}{T_w - 1} & t < T_w \\ \eta_{\min} + \dfrac{1}{2}\left(\eta_{\max} - \eta_{\min}\right)\left(1 + \cos\dfrac{\pi(t - T_w)}{T - T_w}\right) & t \geq T_w \end{cases}$$

Regularization. Stochastic depth (`drop_path` = 0.1): block k of 12 skips its residual branch with probability $0.1 \cdot \frac{k-1}{11}$ during training. LayerScale initialised at 0.1. Dropout and attention dropout are 0. No EMA of the weights (`model_ema` = false).

Model selection and the recording prediction. After every epoch the model scores val and test. A recording's prediction is the mean of its window predictions, and the checkpoint kept (`checkpoint-best.pth`) is the epoch with the lowest val recording-level MAE:

$$\hat{y}_r = \frac{1}{W_r} \sum_{w=1}^{W_r} \hat{y}_{r,w}$$

$$e^* = \underset{e \in \{1, \ldots, 4\}}{\arg\min} \ \mathrm{MAE}_{\mathrm{val}}^{\mathrm{rec}}(e), \qquad \mathrm{MAE}_{\mathrm{val}}^{\mathrm{rec}} = \frac{1}{R} \sum_{r=1}^{R} |\hat{y}_r - y_r|$$

where R is the number of val recordings. The numbers that follow (Tables 0.2 and 0.3) are read from the run itself.

Table 0.2. Training configuration of the run

Generated from the run's `run_config.yaml`; the `Eq.` column names the equation each setting enters.

group	config key	value	meaning	Eq.
data	data.dataset	TUAB_AGE	task: age regression on TUAB	
	data.trim_start_sec	60.0	seconds dropped at the start of every recording	
	data.trim_end_sec	60.0	seconds dropped at the end of every recording	
	data.window_sec	10	window (sample) length, s	
	data.eval_minutes	5.0	val/test budget per recording, minutes (W windows)	(0.10)
	data.case_filter	all	recordings used (normal / abnormal / all)	
	data.random_crop	False	random sub-window crops for training	
input	labram_plus.common_average_reference	True	common average reference	(0.1)
	labram_plus.z_score_patches	False	per-patch z-scoring	
model	model.model	labram_base_patch200_200	backbone (LaBraM-Base, 1 s patches)	
	finetune_checkpoint.finetune	./checkpoints/labram-base.pth	pre-trained weights	
	model.use_mean_pooling	True	mean-pool patch tokens before the head	
	model.init_scale	0.001	head weight scale at init	
	model.drop_path	0.1	max stochastic-depth rate	
	model.layer_scale_init_value	0.1	LayerScale init	
	model.drop	0.0	dropout	
target/loss	loss.regression_loss	huber	regression loss	(0.5)
	loss.huber_delta	1.0	Huber δ (z-score units)	(0.5)
	loss.balance	none	age-balanced weights (B1/B2 runs)	
mixup	mixup.enabled	True	mixup on	(0.4)
	mixup.alpha	0.4	Beta(α, α) parameter	(0.4)
	mixup.prob	1.0	share of batches mixed	(0.4)
	mixup.sigma	0.0	C-Mixup bandwidth (0 = plain mixup)	(0.4)
optimizer	optimizer.opt	adamw	optimizer	(0.7)
	optimizer.weight_decay	0.05	decoupled weight decay λ_wd	(0.7)
	optimizer.layer_decay	0.65	layer-wise LR decay γ	(0.8)
	optimizer.clip_grad	None	gradient clipping	
	trainer.batch_size	64	batch size B	(0.6)
schedule	optimizer.lr	0.0001	peak learning rate η_max	(0.9)
	optimizer.min_lr	1e-06	final learning rate η_min	(0.9)
	optimizer.warmup_lr	1e-06	unused: the warmup starts at 0	(0.9)
	optimizer.warmup_epochs	1	warmup epochs	(0.9)
	trainer.epochs	4	epochs	(0.9)
selection	evaluation.agg_windows	mean	window pooling per recording	(0.10)
	evaluation.use_ema	False	evaluate EMA weights	

Table 0.3. Derived quantities of the run

Values computed from the run (target statistics, step counts, layer-wise LR scales, the selected epoch).

quantity	value	Eq.
μ_y (train-window mean age)	48.88 y	(0.2)
σ_y (train-window age std)	17.82 y	(0.2)
Huber quadratic zone δ·σ_y	±17.8 y	(0.5)
train windows per epoch	268,480	
optimizer steps per epoch / T_w / T	4,195 / 4,195 / 16,780	(0.9)
LR scale: head / last block / first block / embeddings	1.0000 / 0.6500 / 0.0057 / 0.0037	(0.8)
selected epoch e*	4 of 4	(0.11)

1. Training dynamics: how fast does it memorize?

Table 1.1 is the run's `log.txt`, one row per epoch; Figure 1.1 plots it. Every column is computed by `labram/train/train_finetune.py` at the end of the epoch. They differ in four ways: **split**, **unit** (z-scored loss vs. years), **level** (window vs. recording) and **mode** (train mode with mixup and stochastic depth vs. eval mode on unmixed data). The train columns are running averages over the epoch's S optimizer steps, with batch s holding windows $i \in \mathrm{B}_s$, $|\mathrm{B}_s| = B$:

$$\text{train_loss} = \frac{1}{S}\sum_{s=1}^{S} \mathrm{L}_s, \qquad \mathrm{L}_s = \frac{1}{B}\sum_{i \in \mathrm{B}_s} \ell_\delta\left(\hat{z}_i - z_i^{'}\right)$$

$$\text{train_mae} = \frac{1}{S}\sum_{s=1}^{S} \frac{1}{B}\sum_{i \in \mathrm{B}_s} \left|\hat{y}_i - y_i^{'}\right|, \qquad y_i^{'} = \mu_y + \sigma_y z_i^{'}$$

Both are taken on the **mixed** batch (Eq. (0.4)), in train mode, while the weights are still changing within the epoch. `train_mae` is in years but scored against the *mixed* age $y^{'}$ of a *blended* window, so it is not comparable with the val/test MAE. The eval-mode train MAE in Figure 1.1(a) (dash-dot line, Eq. (1.5) on the train recordings, Table 2.2) is the comparable one.

The val (and test) columns are computed in eval mode on unmixed windows, once per epoch, over the N_w windows and R recordings of the split:

$$\text{val_loss} = \frac{1}{S_{\text{val}}}\sum_{s=1}^{S_{\text{val}}} \frac{1}{|\mathrm{B}_s|}\sum_{(r,w) \in \mathrm{B}_s} \ell_\delta\left(\frac{\hat{y}_{r,w} - y_r}{\sigma_y}\right)$$

$$\text{val_window_mae} = \frac{1}{N_w}\sum_{r}\sum_{w=1}^{W_r} \left|\hat{y}_{r,w} - y_r\right|$$

$$\text{val_mae} = \frac{1}{R}\sum_{r=1}^{R} \left|\hat{y}_r - y_r\right|, \qquad \hat{y}_r \text{ from Eq. (0.10)}$$

`test_mae` is Eq. (1.5) on the test recordings. `val_mae` is the model-selection metric (Eq. (0.11)) and the number reported in ClearML. Pooling the 30 windows first averages out window-level noise, which is why `val_mae` < `val_window_mae`.

$$\text{val_age_bias_slope} = \frac{\sum_r (y_r - \bar{y})(e_r - \bar{e})}{\sum_r (y_r - \bar{y})^2}, \qquad e_r = \hat{y}_r - y_r$$

This is the least-squares slope of the recording error on the true age (`utils/regression_metrics.py`). A value of -0.4 means that each extra year of true age adds only 0.6 years of prediction: old recordings are predicted too young and young ones too old (regression to the mean, section 5). The window-level slope is identical, since every recording contributes the same number of windows at one age.

$$\text{val_normal_mae} = \frac{1}{R_{\text{n}}}\sum_{r \in \text{normal}} |\hat{y}_r - y_r|, \qquad \text{val_abnormal_mae} = \frac{1}{R_{\text{a}}}\sum_{r \in \text{abnormal}} |\hat{y}_r - y_r|$$

This is Eq. (1.5) restricted to TUAB's normal or abnormal recordings ($R_{\text{n}} + R_{\text{a}} = R$).

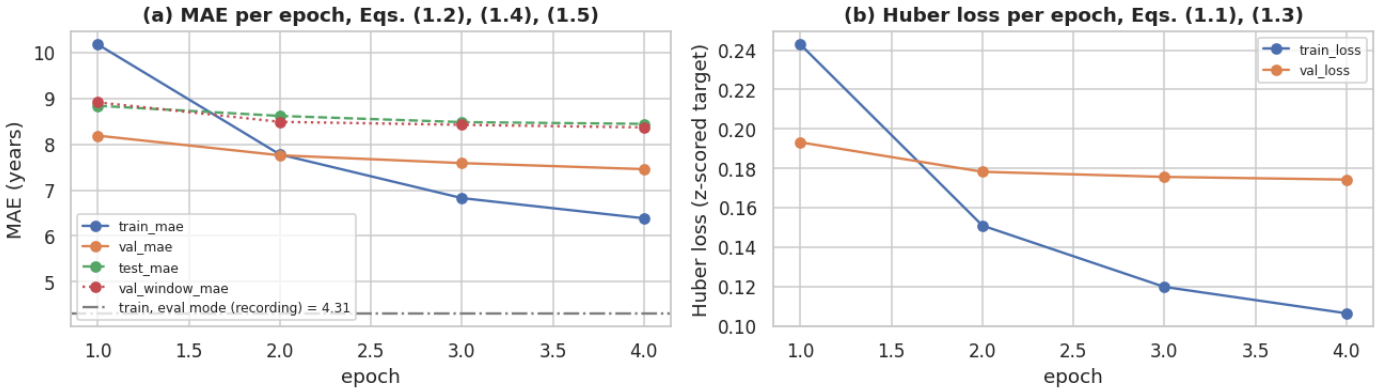
Table 1.1. Per-epoch training log

One row per epoch of `log.txt`; columns defined by Eqs. (1.1)–(1.7).

epoch	train_loss	train_mae	val_loss	val_mae	val_window_mae	test_mae	val_age_bias_slope	val_normal_mae	val_abnormal_mae
1	0.24	10.18	0.19	8.19	8.92	8.84	-0.45	7.49	8.95
2	0.15	7.78	0.18	7.76	8.49	8.62	-0.42	6.89	8.71
3	0.12	6.83	0.18	7.59	8.43	8.48	-0.39	6.66	8.61
4	0.11	6.38	0.17	7.46	8.37	8.45	-0.37	6.56	8.45

Figure 1.1. Training dynamics

(a) MAE per epoch in years: Eqs. (1.2), (1.4), (1.5), and test; the dash-dot line is the eval-mode train recording MAE of the selected checkpoint. (b) Huber loss per epoch, Eqs. (1.1) and (1.3), in z-score units.



2. Error summary: mean vs. median, window vs. recording, cohort

Table 2.2 scores every split at two **levels** and for three **cohorts**:

- **window**: S = all scored windows; each 10 s window is a sample with prediction $\hat{y}_{r,w}$ and target y_r.
- **recording**: S = the recordings; the prediction is the mean over windows, $\hat{y}_r$ (Eq. (0.10)), the reported metric.
- **cohort**: `all`, or only TUAB's `normal` / `abnormal` recordings.

The train split is scored in eval mode on `TRAIN_WINDOWS_PER_RECORDING` windows per recording. Every column is computed by `labram/eval/age_analysis.py::regression_summary` over the n items $i \in \mathrm{S}$ with error $e_i = \hat{y}_i - y_i$ (years); Table 2.1 maps each column to its equation.

$$\mathrm{mae} = \frac{1}{n} \sum_{i \in \mathrm{S}} |e_i|$$

$$\text{median_ae} = \text{median}_{i \in \text{S}} |e_i|$$

$$\text{p90_ae} = Q_{0.90}\left(\{|e_i|\}_{i \in \text{S}}\right)$$

$$\text{rmse} = \sqrt{\frac{1}{n}\sum_{i \in \text{S}} e_i^2}$$

$$\text{r2} = R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i (y_i - \bar{y})^2}$$

$$\text{pearson_r} = \frac{\sum_i (\hat{y}_i - \hat{y})(y_i - \bar{y})}{\sqrt{\sum_i (\hat{y}_i - \hat{y})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}$$

$$\text{mean_err} = \bar{e} = \frac{1}{n}\sum_{i \in \text{S}} e_i$$

$$\text{median_err} = \text{median}_{i \in \text{S}} e_i$$

$$\text{bias_slope} = b = \frac{\sum_i (y_i - \bar{y})(e_i - \bar{e})}{\sum_i (y_i - \bar{y})^2} = a - 1, \qquad \hat{y}_i \approx a\, y_i + c \text{ (least squares)}$$

Here $Q_{0.90}$ is the 90th percentile (linear interpolation) and $\bar{y}$, $\hat{y}$ are the means over S.

Table 2.1. Column key for Table 2.2

column	Eq.	what it tells
`n`	–	windows or recordings in the row
`mae`	(2.1)	average error size in years; the headline metric (recording level = `val_mae`, Eq. (1.5))
`median_ae`	(2.2)	the typical error; less sensitive than `mae` to the large-error tail
`p90_ae`	(2.3)	90% of the items are within this many years
`rmse`	(2.4)	weights large errors quadratically; `rmse` ≫ `mae` means a heavy tail
`r2`	(2.5)	share of the age variance explained; 0 = always predicting $\bar{y}$; can be negative. Not the square of `pearson_r` unless the fit is unbiased with slope 1
`pearson_r`	(2.6)	linear association of prediction and age; ignores offset and scale
`mean_err`	(2.7)	signed bias: > 0 predicted too old on average
`median_err`	(2.8)	signed bias of the typical item
`bias_slope`	(2.9)	slope of the error against age; $b = -0.4$ means $\hat{y} \approx 0.6\,y + c$, so a 10-year age difference shows up as a 6-year prediction difference (regression to the mean, section 5)

`bias_slope` is the same at window and recording level (each recording has one age and, on val/test, the same number of windows). `mae`, `rmse` and `p90_ae` shrink from window to recording level because pooling averages out window noise.

Table 2.2. Error summary per level, split and cohort

Years; columns as in Table 2.1, Eqs. (2.1)–(2.9).

level	split	cohort	n	mae	median_ae	p90_ae	rmse	r2	pearson_r	mean_err	median_err	bias_slope
		all	16680	8.37	6.78	17.74	10.74	0.54	0.75	-0.40	-0.26	-0.37
	val	abnormal	7950	9.37	7.42	20.27	12.01	0.38	0.65	-0.54	-0.33	-0.46
		normal	8730	7.46	6.26	15.63	9.44	0.62	0.79	-0.28	-0.21	-0.34
		all	8220	9.28	7.27	19.63	12.17	0.55	0.74	-1.40	-0.75	-0.41
window	test	abnormal	3750	11.40	9.43	23.20	14.56	0.30	0.59	-2.35	-2.86	-0.56
		normal	4470	7.49	5.99	15.47	9.72	0.65	0.81	-0.60	0.49	-0.35
		all	21450	5.49	4.28	11.57	7.37	0.83	0.91	-0.62	-0.30	-0.20
	train	abnormal	10700	5.19	4.05	11.03	6.96	0.84	0.92	-0.65	-0.45	-0.18
		normal	10750	5.79	4.55	12.04	7.75	0.77	0.88	-0.58	-0.12	-0.26
		all	556	7.46	6.14	16.08	9.64	0.63	0.80	-0.40	-0.09	-0.37
	val	abnormal	265	8.45	6.64	18.53	10.93	0.48	0.70	-0.54	-0.20	-0.46
		normal	291	6.56	5.78	13.08	8.30	0.71	0.84	-0.28	-0.02	-0.34
		all	274	8.45	6.78	17.72	11.20	0.62	0.79	-1.40	-0.34	-0.41
recording	test	abnormal	125	10.65	8.79	22.36	13.65	0.39	0.64	-2.35	-2.88	-0.56
		normal	149	6.60	5.27	13.40	8.63	0.73	0.86	-0.60	0.89	-0.35
		all	2145	4.31	3.35	9.00	5.89	0.89	0.95	-0.62	-0.21	-0.20
	train	abnormal	1070	3.95	2.99	8.31	5.39	0.91	0.96	-0.65	-0.38	-0.18
		normal	1075	4.67	3.72	9.43	6.35	0.85	0.93	-0.58	-0.01	-0.26

3. Error and loss distributions

All ages are in years, on the original scale. Figure 3.1 has six panels:

- (a) the distribution of **true vs. predicted age** per recording. The predicted distribution is visibly narrower, which is regression to the mean.
- (b) the **predicted age for each true-age decade**: the box spans the IQR, the line is the median and the whiskers reach 1.5 IQR. The green band marks where the prediction would equal the true age.
- (c) the empirical CDF of the recording absolute error,

$$\hat{F}(a) = \frac{1}{n} \#\{\, r : |e_r| \le a \,\}$$

- (d) the recording $|e_r|$ by cohort.
- (e) and (f) the per-window **Huber loss** in eval mode, Eq. (0.5) on the unmixed window, $\ell_{r,w} = \ell_\delta\big((\hat{y}_{r,w} - y_r)/\sigma_y\big)$, with $\delta = 1$. Panel (e) shows how much of the total loss sits in the worst windows. With the per-window losses sorted in decreasing order, $\ell_{(1)} \ge \ell_{(2)} \ge \cdots \ge \ell_{(N)}$:

$$\text{share}(q) = \frac{\sum_{k=1}^{\lceil qN \rceil} \ell_{(k)}}{\sum_{k=1}^{N} \ell_{(k)}}, \qquad q \in [0, 1]$$

Table 3.1 gives quantiles of the recording $|e_r|$ (Q_p as in Eq. (2.3)).

Figure 3.1. Error and loss distributions

Val/test recordings and windows; panels (a)–(f) as listed above, Eqs. (3.1) and (3.2).

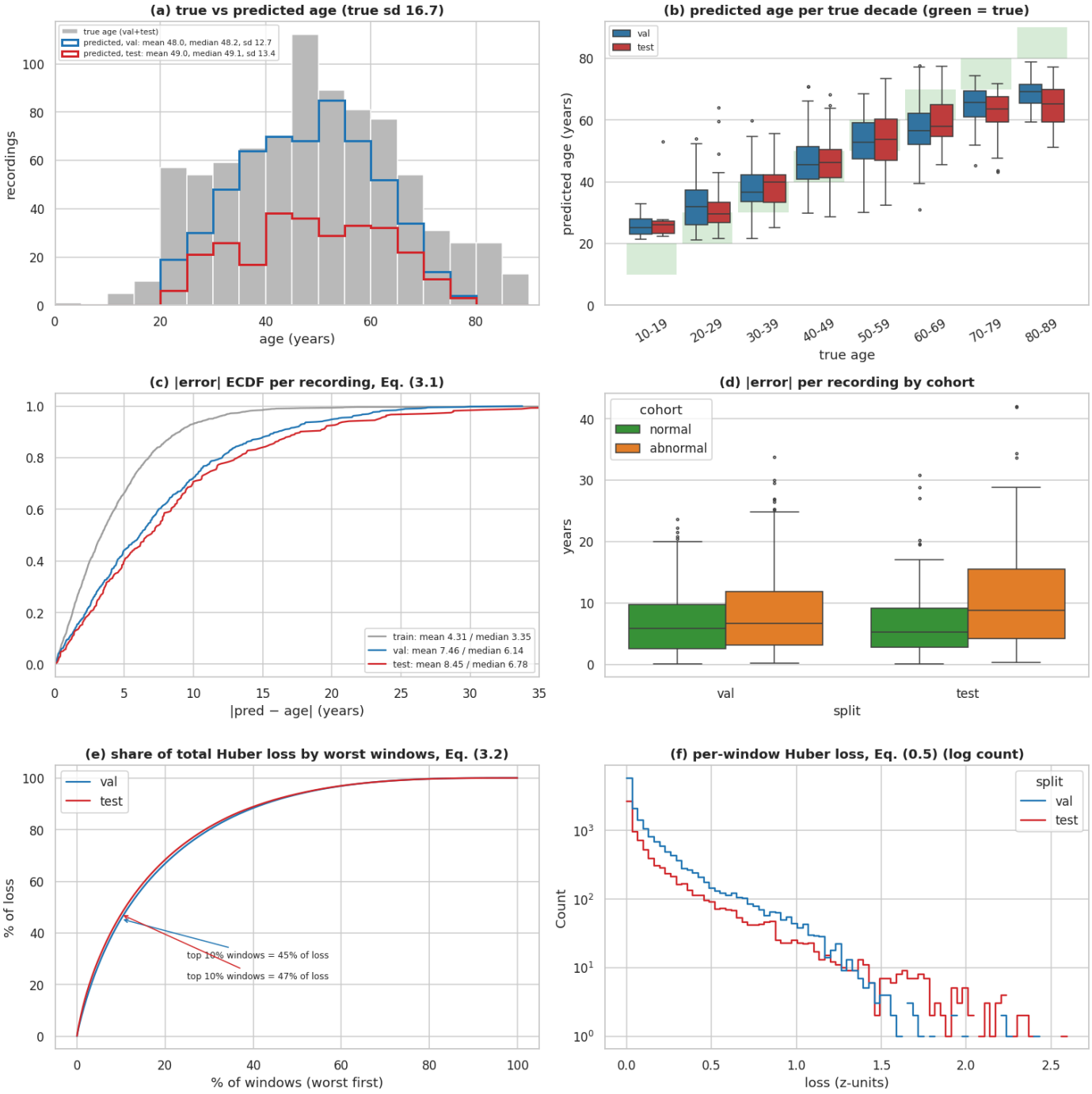


Table 3.1. Quantiles of the recording absolute error

Mean and quantiles Q_p (Eq. (2.3)) of $|e_r|$ in years, val and test.

split	|err| mean	|err| min	|err| 50%	|err| 75%	|err| 90%	|err| 95%	|err| max
test	8.45	0.04	6.78	11.39	17.72	23.37	42.07
val	7.46	0.00	6.14	10.51	16.08	20.21	33.77

4. Per-sample view: windows inside a recording

How much do a recording's 30 window predictions disagree? Does the disagreement flag a bad recording? Does median pooling beat mean pooling? Does the position in the recording matter? TUAB sessions often contain hyperventilation and photic stimulation, and patients get drowsy as a session goes on.

The within-recording spread is the sample std of the window predictions; median pooling replaces the mean of Eq. (0.10) with the median; the window's position is its start time $t_w = 10\,w/60$ minutes from the start of the EDF, with w the window number in the file name:

$$s_r = \sqrt{\frac{1}{W_r - 1} \sum_{w=1}^{W_r} \left(\hat{y}_{r,w} - \hat{y}_r\right)^2}$$

$$\hat{y}_r^{\text{med}} = \text{median}_w\, \hat{y}_{r,w}, \qquad e_r^{\text{med}} = \hat{y}_r^{\text{med}} - y_r$$

Figure 4.1. Window predictions inside a recording

(a) Histogram of s_r (Eq. (4.1)). (b) s_r against the recording $|e_r|$, with Pearson r (Eq. (2.6)). (c) Window MAE (Eq. (2.1)) and median AE (Eq. (2.2)) per minute of recording; bins with at least 100 windows. (d) Recording $|e_r|$ under mean vs. median pooling (Eq. (4.2)); points below the diagonal are better with the median.

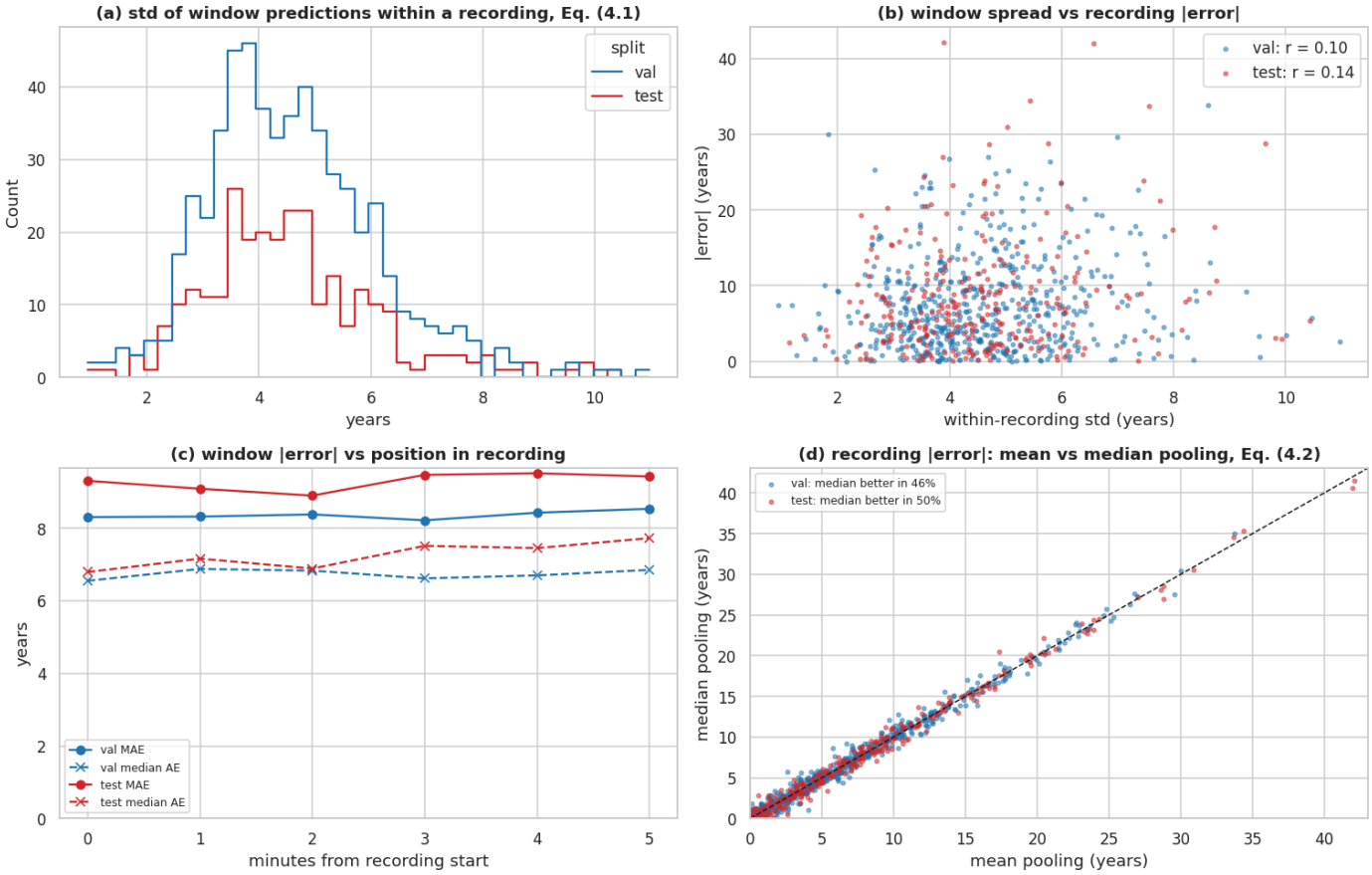


Table 4.1. MAE by pooling

Window MAE (Eq. (2.1)), recording MAE with mean (Eq. (0.10)) and median (Eq. (4.2)) pooling, and the mean within-recording spread s_r (Eq. (4.1)), per split.

split	window MAE	recording MAE (mean pool)	recording MAE (median pool)	mean within-rec std
train	5.49	4.31	4.24	4.36
val	8.37	7.46	7.53	4.58
test	9.28	8.45	8.43	4.58

5. Regression to the mean, against the train sampling density

The core failure. Young subjects are predicted older and old subjects younger than they are. Per split, a least-squares line of prediction on age,

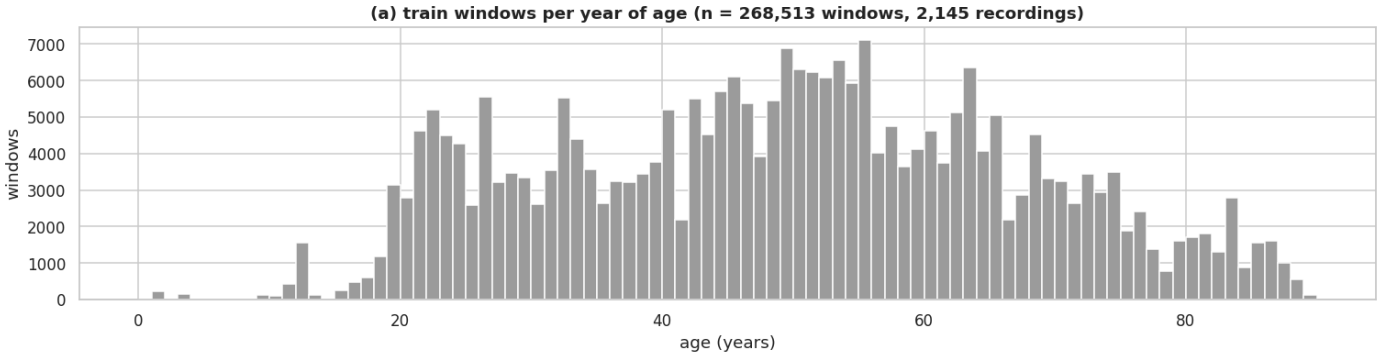
$$\hat{y}_r \approx a\, y_r + c, \qquad a = 1 + b$$

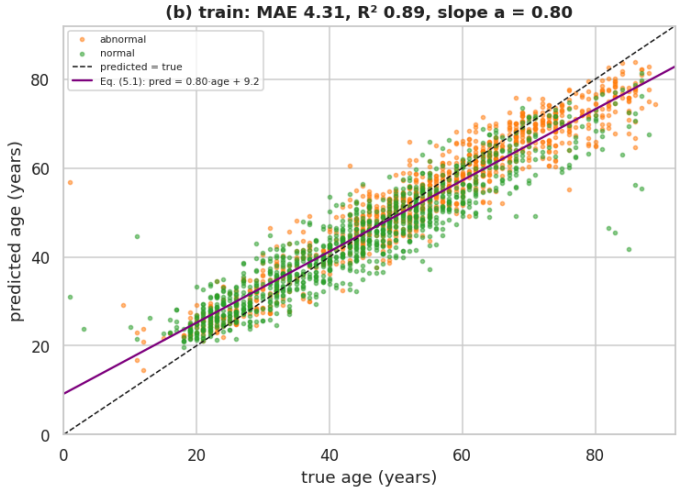
has slope $a < 1$, with b the `bias_slope` of Eq. (2.9). Each true-age decade d (bins $[0, 10)$, $[10, 20)$, …, $[80, 90)$) is summarised over its recordings $r \in d$, and set against the number of **train windows** in the decade:

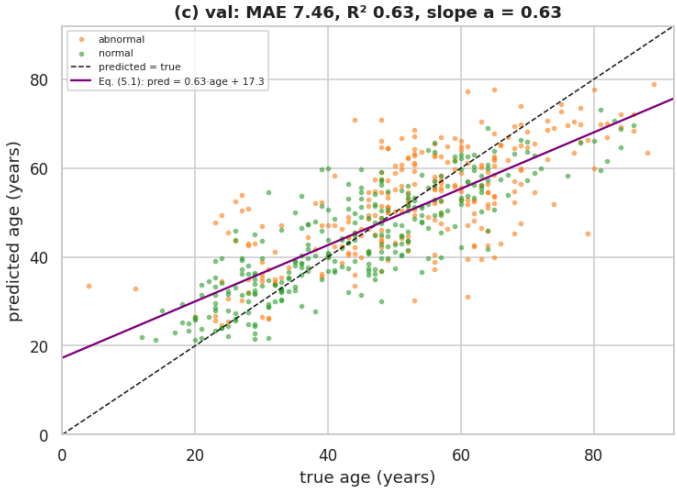
$$\bar{\hat{y}}_d = \frac{1}{n_d}\sum_{r\in d}\hat{y}_r, \qquad \mathrm{MAE}_d = \frac{1}{n_d}\sum_{r\in d}|e_r|, \qquad N_d^{\mathrm{train}} = \#\{\text{train windows } i : y_i \in d\}$$

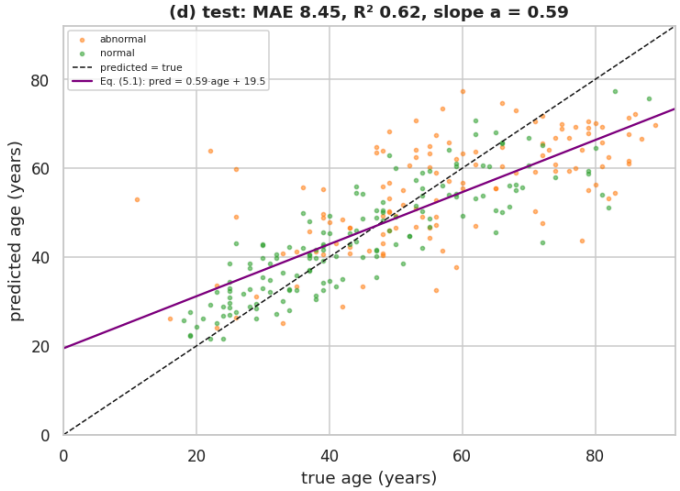
Figure 5.1. Regression to the mean vs. train sampling density

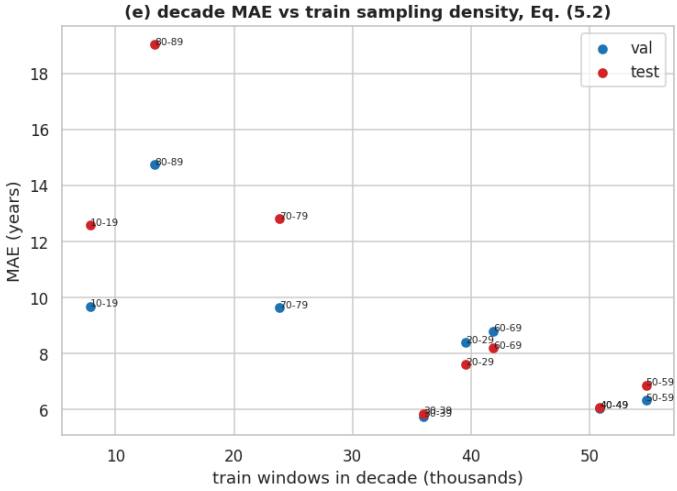
(a) Train windows per year of age: the model sees ages 40–60 far more often than children or people over 80. (b)–(d) Predicted vs. true age per recording for train (eval mode), val and test, coloured by cohort, with the fit of Eq. (5.1). (e) Decade MAE (Eq. (5.2)) against the decade's train density N_d^{train}. If the bias simply followed sampling density, the worst-served decades would be the rarest ones. (f) Median predicted age with its IQR per true-age decade; the further it sits from the diagonal, the stronger the pull toward the train mean μ_y (~49 y).


(a) train windows per year of age (n = 268,513 windows, 2,145 recordings)
windows
age (years)


(b) train: MAE 4.31, R² 0.89, slope a = 0.80
abnormal
normal
predicted = true
Eq. (5.1): pred = 0.80·age + 9.2
predicted age (years)
true age (years)


(c) val: MAE 7.46, R² 0.63, slope a = 0.63
abnormal
normal
predicted = true
Eq. (5.1): pred = 0.63·age + 17.3
predicted age (years)
true age (years)


(d) test: MAE 8.45, R² 0.62, slope a = 0.59
abnormal
normal
predicted = true
Eq. (5.1): pred = 0.59·age + 19.5
predicted age (years)
true age (years)


(e) decade MAE vs train sampling density, Eq. (5.2)
val
test
MAE (years)
train windows in decade (thousands)

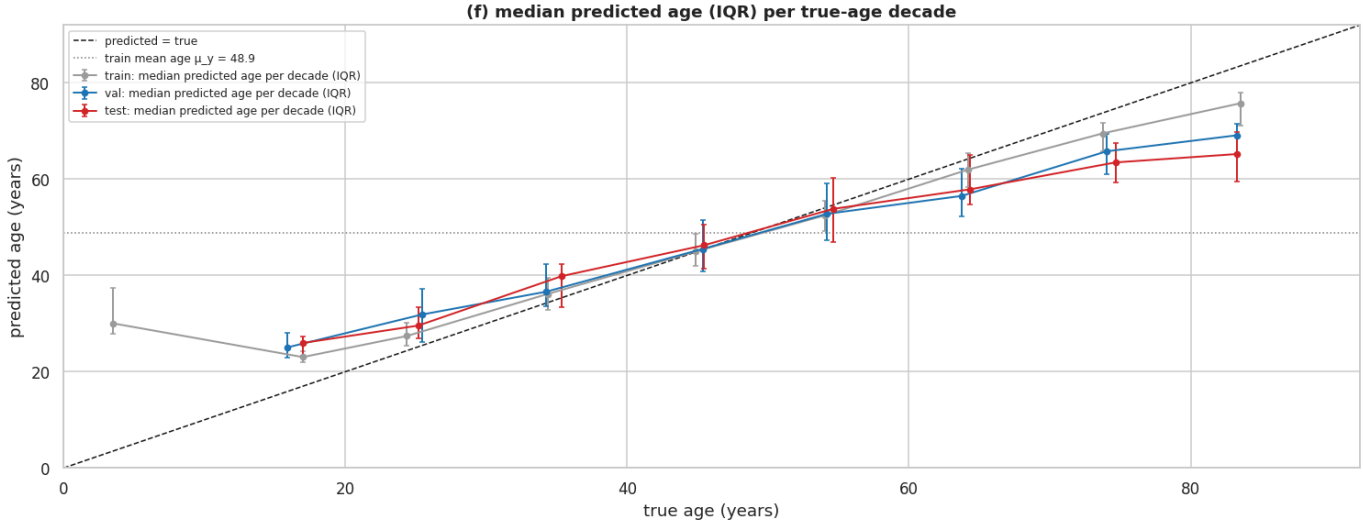

(f) median predicted age (IQR) per true-age decade
predicted = true
train mean age μ_y = 48.9
train: median predicted age per decade (IQR)
val: median predicted age per decade (IQR)
test: median predicted age per decade (IQR)
predicted age (years)
true age (years)

Table 5.1. Error per true-age decade

Per decade and split: `n` = n_d; `age_mean`; `pred_mean` = $\bar{\hat{y}}_d$; `pred_median`, `pred_q25`, `pred_q75` = the median and quartiles of $\hat{y}_r$ in the decade; `mae` = MAE_d; `median_ae` = Eq. (2.2) within the decade.

split	age_bin	n	age_mean	pred_mean	pred_median	pred_q25	pred_q75	mae	median_ae
test	**10-19**	6	17.00	29.58	25.98	23.31	27.30	12.58	8.23
	20-29	34	25.18	32.42	29.58	26.84	33.42	7.62	4.74
	30-39	49	35.37	39.01	39.86	33.33	42.36	5.85	4.84
	40-49	52	45.48	47.07	46.28	41.41	50.49	6.07	4.90
	50-59	51	54.65	53.73	53.84	47.01	60.32	6.87	6.18
	60-69	32	64.31	59.47	57.87	54.75	64.99	8.19	8.83
	70-79	30	74.67	61.85	63.48	59.28	67.55	12.82	10.60
	80-89	20	83.30	64.26	65.22	59.45	69.84	19.04	19.38
	0-9	1	4.00	33.57	33.57	33.57	33.57	29.57	29.57
val	**10-19**	9	15.89	25.58	25.05	22.99	28.01	9.69	7.36
	20-29	77	25.43	33.04	31.87	26.18	37.22	8.40	6.53
	30-39	75	34.25	38.10	36.62	33.51	42.27	5.77	3.76
	40-49	130	45.35	46.80	45.50	40.83	51.48	6.05	4.90
	50-59	119	54.19	52.94	52.80	47.38	59.08	6.33	6.15
	60-69	99	63.75	56.52	56.52	52.19	62.24	8.80	7.20
	70-79	27	74.04	64.40	65.78	60.94	69.35	9.64	8.11
	80-89	19	83.32	68.57	69.12	65.44	71.46	14.75	14.88

6. Post-hoc bias correction (fit on val, apply to test)

Each correction is fitted on the validation recordings (all, or normal only, which is the usual "healthy controls" choice) and applied to test. Two least-squares fits on val (`fit_bias_correction`):

$$\hat{y} \approx a\,y + c \qquad \text{and} \qquad y \approx \alpha\,\hat{y} + \beta$$

The corrections (`apply_bias_correction`):

$$\textbf{cole:} \quad \hat{y}' = \frac{\hat{y} - c}{a}$$

$$\textbf{recalibrate:} \quad \hat{y}' = \alpha\,\hat{y} + \beta$$

$$\textbf{age_level:} \quad \hat{y}' = \hat{y} - (a - 1)\,y - c$$

- **cole** inverts the fit of Eq. (6.1). It is deployable, since it uses only the prediction, and it sets the bias slope (Eq. (2.9)) to about 0.
- **recalibrate** is the MSE-optimal linear map from prediction to age. It is deployable, and with $\alpha \approx 1$ it changes little.
- **age_level** subtracts the age-dependent bias using the **true age** y. This is the standard brain-age-*gap* correction. It is not an age estimate, and its MAE cannot be compared with the others.

Table 6.1. Test metrics after each bias correction

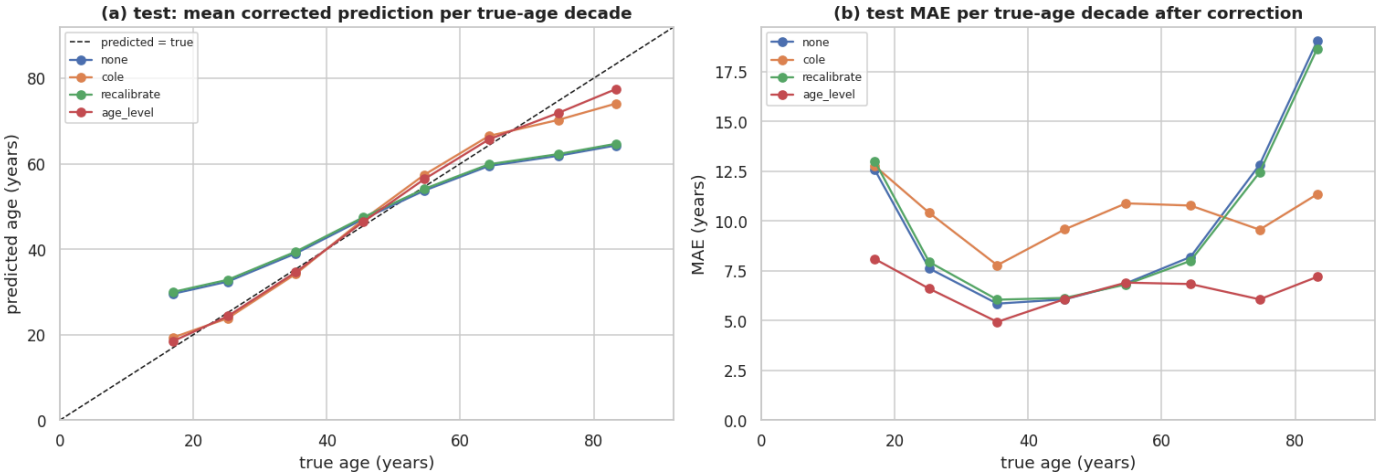
Test metrics after each correction: MAE (Eq. (2.1)), median AE (Eq. (2.2)), bias slope (Eq. (2.9)), and MAE per cohort (Eq. (1.7)).

`fit on val: pred = 0.634 * age + 17.29` | `age ~= 0.999 * pred + 0.46`

fit on	method	deployable	test MAE	median AE	bias slope	normal MAE	abnormal MAE
val (all)	none	True	8.45	6.78	-0.41	6.60	10.65
	cole	True	9.94	8.45	-0.08	8.40	11.78
	recalibrate	True	8.44	6.61	-0.41	6.64	10.59
	age_level	False	6.30	5.36	-0.05	5.33	7.47
val (normal)	none	True	8.45	6.78	-0.41	6.60	10.65
	cole	True	9.55	6.89	-0.11	7.73	11.72
	recalibrate	True	8.34	6.77	-0.37	6.51	10.53
	age_level	False	6.30	4.55	-0.07	5.10	7.73

Figure 6.1. Post-hoc bias corrections on test

Test, per true-age decade (Eq. (5.2)): (a) mean corrected prediction; (b) MAE after correction.



7. Normal vs. abnormal, and the spectral correlates of the error

The abnormal recordings are older and their EEG is slower: more delta/theta, less alpha and a lower peak alpha frequency (PAF). EEG slowing is also a classic marker of ageing. The question is whether the model mistakes pathological slowing for age.

Features per recording (`recording_spectral_features`) are computed after CAR (Eq. (0.1), without the /100), over up to 6 of its scored windows. For channel c of window w, $P_{c,w}(f)$ is the Welch PSD (2 s Hann segments, 50% overlap, 0.5 Hz resolution, µV²/Hz). For a band $\mathrm{B} \in \{\delta\,[1,4),\ \theta\,[4,8),\ \alpha\,[8,13),\ \beta\,[13,30),\ \gamma\,[30,45)\}$ Hz:

$$\mathrm{rel_B} = \frac{1}{|w|\,C} \sum_{w} \sum_{c} \frac{\sum_{f \in \mathrm{B}} P_{c,w}(f)}{\sum_{1 \le f < 45} P_{c,w}(f)}$$

$$\mathrm{paf} = \arg\max_{7 \le f \le 13} \frac{1}{|\mathrm{P}|} \sum_{c \in \mathrm{P}} \frac{1}{|w|} \sum_{w} P_{c,w}(f), \qquad \mathrm{P} = \{\mathrm{O1}, \mathrm{O2}, \mathrm{P3}, \mathrm{P4}, \mathrm{Pz}, \mathrm{T5}, \mathrm{T6}\}$$

$$\mathrm{theta_alpha} = \frac{\mathrm{rel_theta}}{\mathrm{rel_alpha}}, \qquad \mathrm{amp_uv} = \frac{1}{|w|\,C} \sum_{w} \sum_{c} \mathrm{std}_t\left(x_{c,t}\right)$$

A PAF of 7 Hz sits at the edge of the search band and means *no alpha peak*, which is typical of diffusely slowed abnormal EEG. The error is **age-detrended** (its least-squares line in age over the val+test recordings is removed), so that regression to the mean does not masquerade as a spectral effect. Associations use Spearman's rank correlation, i.e. Eq. (2.6) applied to the ranks:

$$\tilde{e}_r = e_r - (k\,y_r + m), \qquad (k, m) = \arg\min_{k,m} \sum_{r} \left(e_r - k\,y_r - m\right)^2$$

$$\rho(u, v) = \mathrm{pearson}\left(\mathrm{rank}(u),\ \mathrm{rank}(v)\right)$$

Table 7.1. Spectral features by cohort

Median per recording and cohort of age, $|e_r|$, e_r and the features of Eqs. (7.1)–(7.3).

cohort median per recording	abnormal	normal
age	55.00	44.00
abs_err	7.42	5.44
err	-0.77	0.06
paf	7.50	10.00
theta_alpha	1.40	0.47
rel_delta	0.53	0.37
rel_theta	0.16	0.11
rel_alpha	0.11	0.24
rel_beta	0.10	0.16
amp_uv	14.91	12.58

Table 7.2. Spectral correlates of age and of the detrended error

Spearman ρ (Eq. (7.5)) of each feature with age and with the detrended error $\tilde{e}_r$ (Eq. (7.4)), overall and within each cohort.

	ρ(feature, age)	ρ(feature, age-detrended error)	ρ within normal	ρ within abnormal
paf	-0.27	-0.22	-0.13	-0.15
theta_alpha	0.17	0.17	-0.02	0.15
rel_delta	0.06	0.09	-0.17	0.15
rel_theta	0.05	0.06	-0.08	0.05
rel_alpha	-0.15	-0.16	-0.00	-0.13
rel_beta	-0.04	-0.07	0.23	-0.17
amp_uv	-0.05	-0.05	-0.15	-0.02

Figure 7.1. Normal vs. abnormal, and the peak alpha frequency

(a) True-age distribution per cohort. (b) Median predicted age (IQR) per decade by cohort. (c) The same by PAF group. (d) True vs. predicted age coloured by PAF.

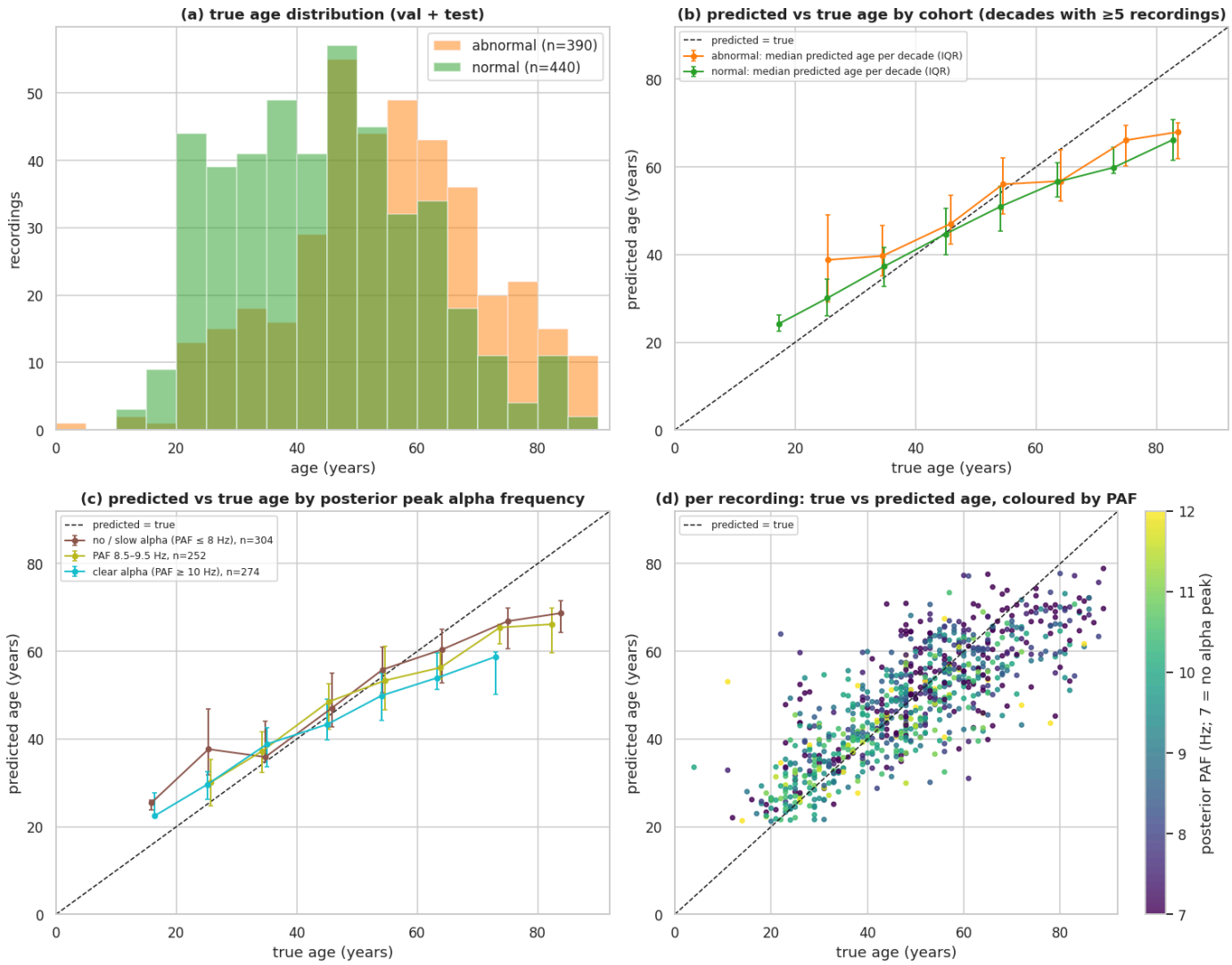


Table 7.3. Recordings by posterior PAF group

Per PAF group: count, abnormal share, mean true and predicted age, median $\tilde{e}_r$ and MAE.

paf_group	n	abnormal_share	true_age	predicted_age	age_detrended_error	mae
≤ 8 Hz	304	0.80	54.51	53.69	1.66	8.77
8.5–9.5 Hz	252	0.38	48.23	47.52	-0.35	7.60
≥ 10 Hz	274	0.19	43.77	43.11	-2.46	6.85

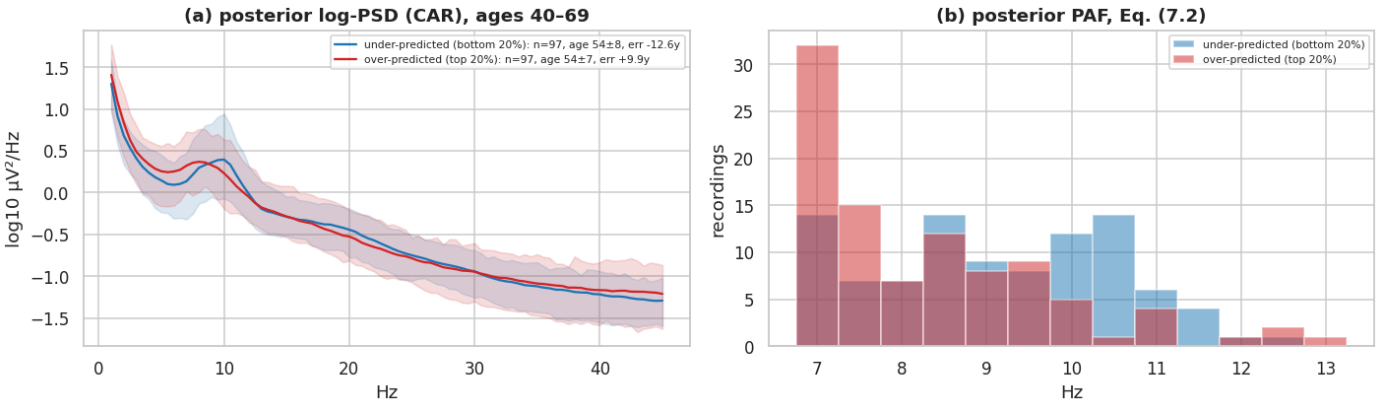
Average spectra of over- vs. under-predicted recordings, age-matched

Recordings in the top and bottom quintiles of the age-detrended error $\tilde{e}_r$ (Eq. (7.4)). Each quintile is compared within the 40–69 age range, so that the groups have similar ages.

Figure 7.2. Spectra of under- vs. over-predicted recordings, ages 40–69

(a) Mean posterior log-PSD, $\log_{10}$ of the posterior-channel average of $P_{c,w}(f)$ over 6 windows per recording, with the IQR across recordings. (b) Histogram of the posterior PAF (Eq. (7.2)) per

group.




```
under-predicted (bottom 20%)     abnormal share 43%, median PAF 9.0 Hz, median theta/al
pha 0.73
over-predicted (top 20%)         abnormal share 72%, median PAF 8.0 Hz, median theta/al
pha 1.25
```

8. Real EEG: accurate vs. too young vs. too old, at the same true age

For each age in `COMPARE_AGES` (Parameters), three val/test recordings of about that true age (±3 y): the one predicted **most accurately**, the one predicted **youngest** (under-predicted) and the one predicted **oldest** (over-predicted). The EEG is real: one 10 s window per recording (the window whose prediction is closest to the recording's mean), common-average referenced as the model sees it, 19 standard 10-20 channels, and the **same µV scale** in all three panels. The figure's lower panels hold the per-window predictions, the posterior spectra and the relative band powers.

Figures 8.1–8.4 (one per age in `COMPARE_AGES`). Panels (a)–(c) show the EEG of the accurate, under- and over-predicted recordings, (d) the per-window predictions across each recording, (e) the posterior spectrum and (f) the relative band power (Eq. (7.1)).

Change `COMPARE_AGES`, or call `ex.compare_at_age(age, tol=3, cohort="normal", split="test")` yourself.

Figure 8.1. Real EEG at true age ≈ 25: accurate vs. under- vs. over-predicted

CAR, same 80 µV scale in (a)–(c), clipped at ±1 trace spacing; the window closest to each recording's mean prediction.

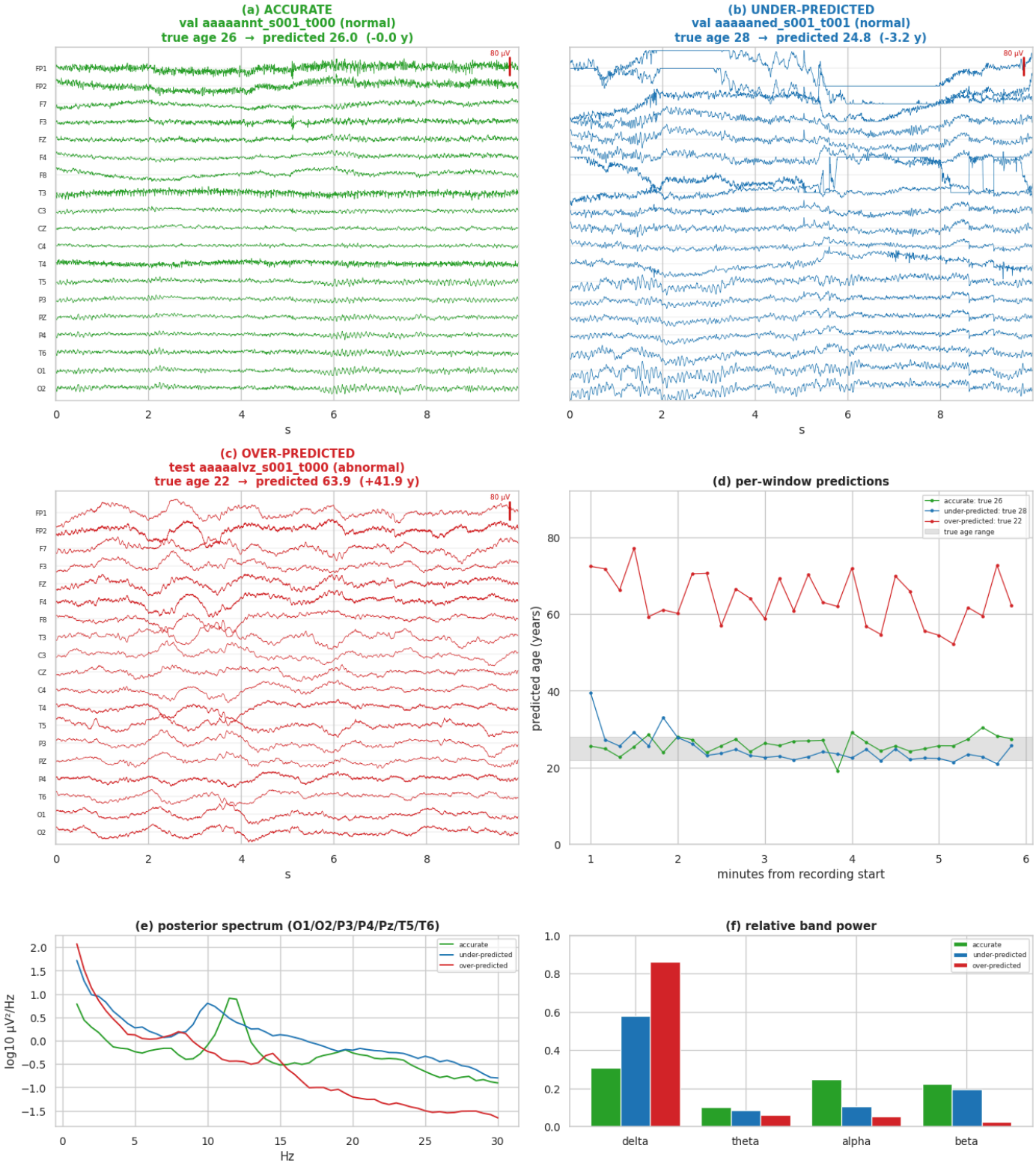


Figure 8.2. Real EEG at true age ≈ 45: accurate vs. under- vs. over-predicted

CAR, same 60 µV scale in (a)–(c), clipped at ±1 trace spacing; the window closest to each recording's mean prediction.

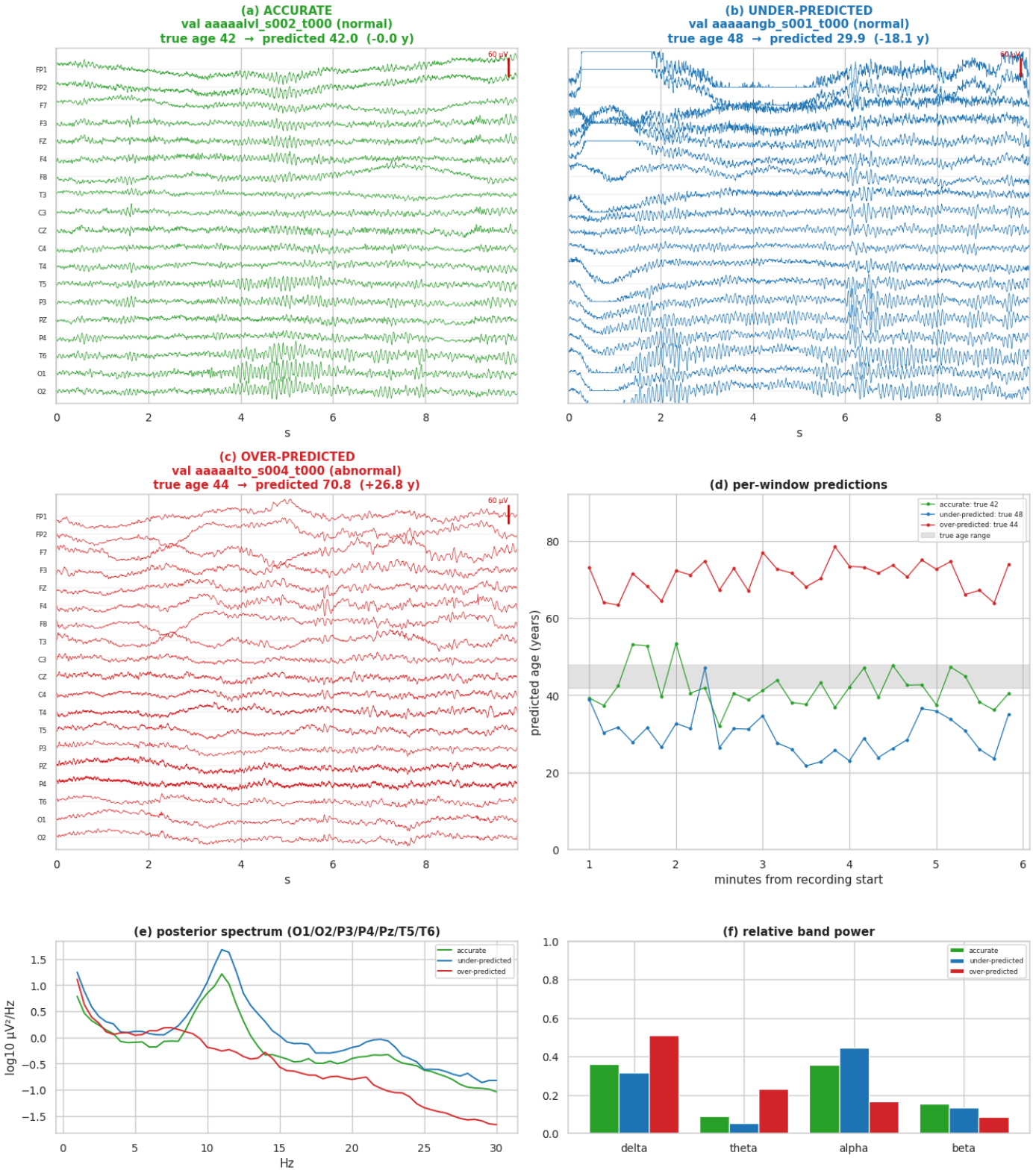


Figure 8.3. Real EEG at true age ≈ 65: accurate vs. under- vs. over-predicted

CAR, same 100 µV scale in (a)–(c), clipped at ±1 trace spacing; the window closest to each recording's mean prediction.

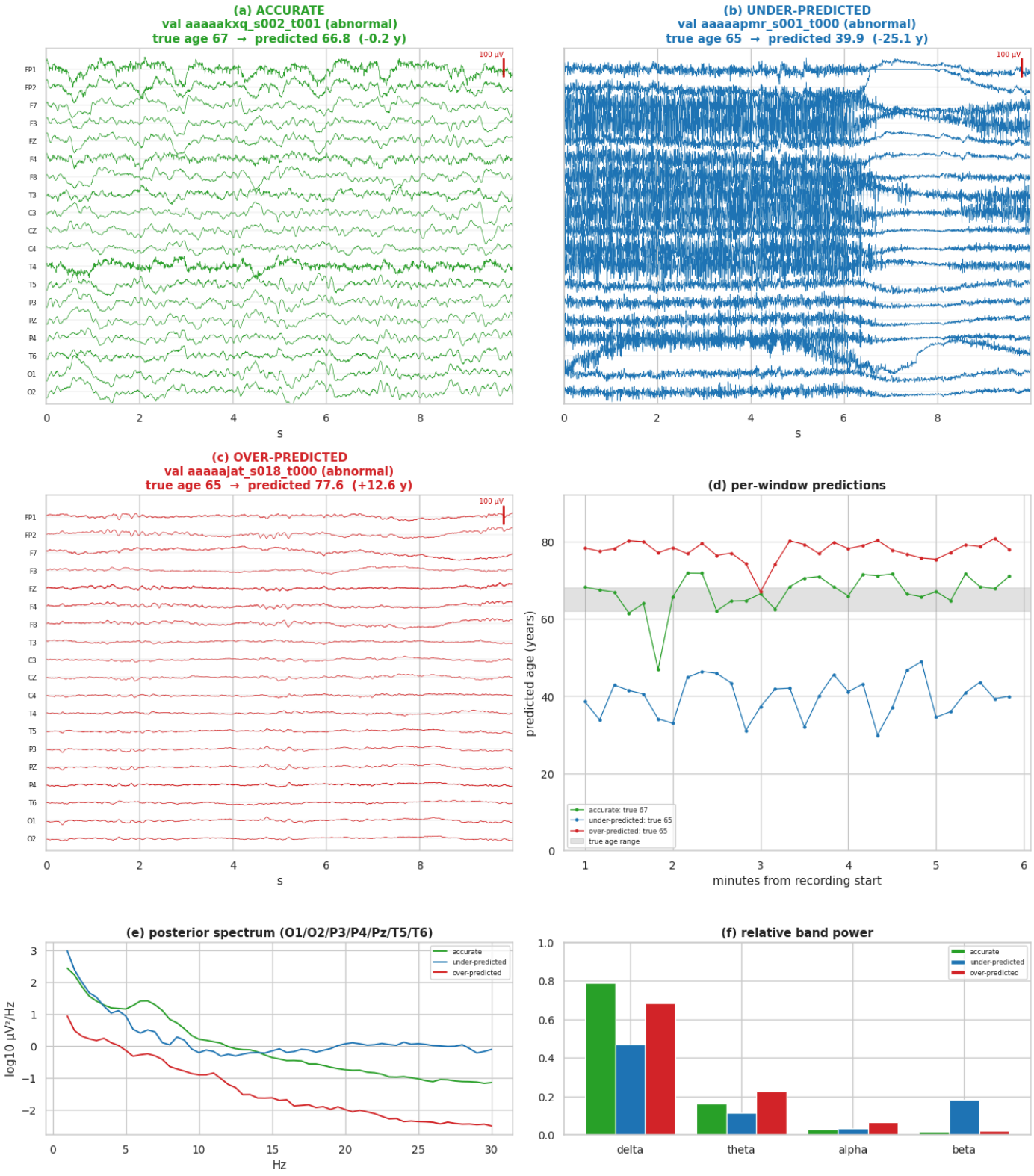
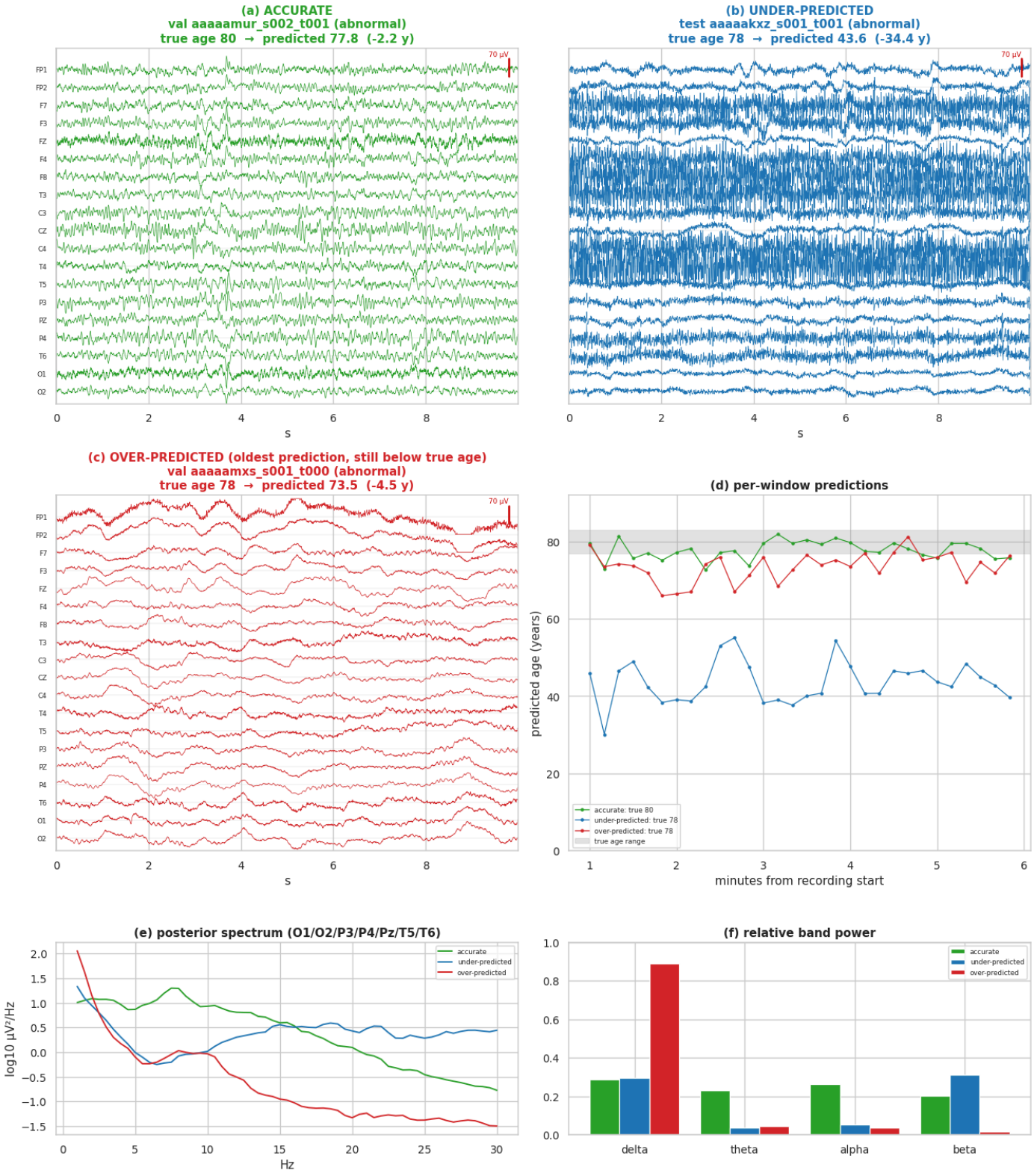


Figure 8.4. Real EEG at true age ≈ 80: accurate vs. under- vs. over-predicted

CAR, same 70 µV scale in (a)–(c), clipped at ±1 trace spacing; the window closest to each recording's mean prediction.



What Figures 8.1–8.4 show (M1, as executed). At the same true age the three outcomes look different on the screen:

- **Too old** (red): slow EEG. Delta/theta dominate (50–89% delta), with a low or absent posterior alpha peak and little beta. Examples: a 22-year-old predicted 64, and a 44-year-old predicted 71 (both abnormal, diffuse slowing).

- **Too young** (blue): *fast* EEG, of two kinds. At 45 it is a normal recording with a strong, high-frequency alpha rhythm (48 → 30). At 65 and 80 it is heavy high-frequency **muscle (EMG) artifact** across most channels, with beta at 17–31% of the power (65 → 40, 78 → 44). The model reads both as a "young" spectrum. Artifact-contaminated recordings are a concrete source of the largest under-predictions. However, re-pooling with EMG-heavy windows dropped or down-weighted (`scripts/age_artifact_pooling.py` ; 30–45 Hz relative power per window) does **not** lower the MAE: on val the best setting for M1 is no rejection, and on test no setting moves the MAE by more than about 0.1 y. The artifact runs through the whole recording, so robust *inputs* (artifact-aware training) are the remaining lever, not pooling.
- **Accurate** (green): an age-typical background, e.g. a clear ~11 Hz alpha at 26 and 42.
- At 80, regression to the mean is visible directly: even the *oldest* prediction near 80 is below the true age (78 → 73.5). At 25, the *youngest* prediction is only 3 years too young (28 → 25), because nearly every young recording is pulled upward.

The per-window traces show that these are consistent offsets across the whole recording, not a few bad windows.

8b. The extreme cases

The two most accurate, most over-predicted and most under-predicted recordings in val and test (any age), listed in Table 8.1. Figures 8.5 onward, one per row of Table 8.1 in that order, each show (a) the recording's most representative window, (b) all 30 window predictions on the age axis, (c) the posterior spectrum against age-matched normal recordings, and (d) the band powers (Eq. (7.1)).

Table 8.1. Extreme cases

The two most accurate, most over- and most under-predicted recordings per split (years); `figure` names the figure that shows each.

	split	kind	recording	cohort	age	pred	err	pred_std	n_windows	figure
0	val	best	aaaaannt_s001_t000	normal	26.00	26.00	-0.00	2.15	30	8.5
1	val	best	aaaaalvl_s002_t000	normal	42.00	41.98	-0.02	5.11	30	8.6
2	val	over-predicted	aaaaancc_s001_t000	abnormal	4.00	33.57	29.57	6.99	30	8.7
3	val	over-predicted	aaaaagxr_s009_t001	abnormal	27.00	53.94	26.94	4.69	30	8.8
4	val	under-predicted	aaaaamtk_s001_t000	abnormal	79.00	45.23	-33.77	8.62	30	8.9
5	val	under-predicted	aaaaampk_s001_t000	abnormal	61.00	31.02	-29.98	1.83	30	8.10
6	test	best	aaaaapof_s001_t000	normal	37.00	37.04	0.04	7.37	30	8.11
7	test	best	aaaaaoeg_s001_t000	normal	43.00	42.94	-0.06	4.06	30	8.12
8	test	over-predicted	aaaaaohb_s001_t000	abnormal	11.00	53.07	42.07	3.89	30	8.13
9	test	over-predicted	aaaaalvz_s001_t000	abnormal	22.00	63.95	41.95	6.58	30	8.14
10	test	under-predicted	aaaaakxz_s001_t001	abnormal	78.00	43.62	-34.38	5.43	30	8.15
11	test	under-predicted	aaaaaiwz_s001_t001	normal	82.00	51.11	-30.89	5.03	30	8.16

Figure 8.5. val aaaaannt_s001_t000 (normal): true age 26, recording prediction 26.0 (-0.0 y)

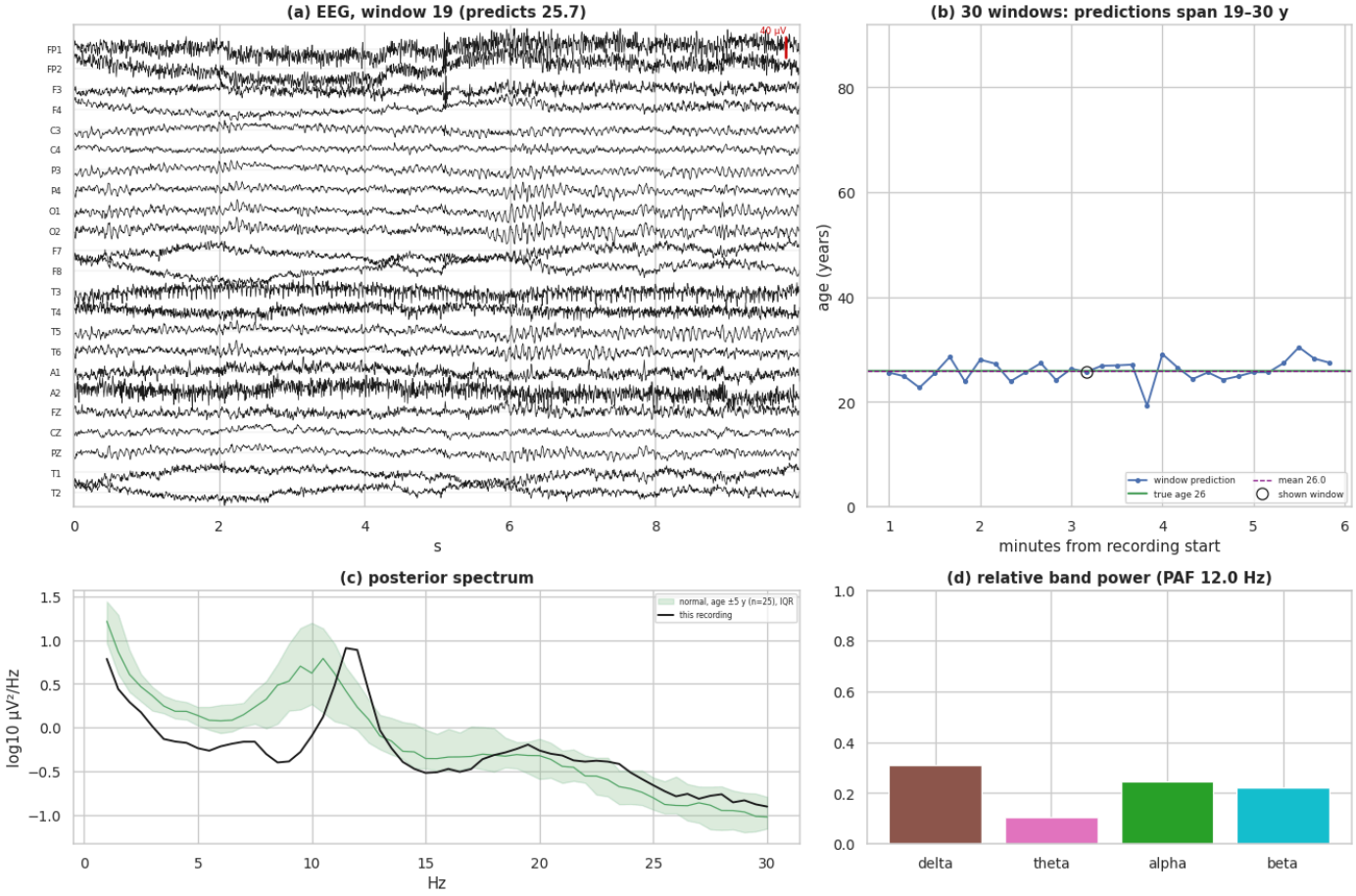


Figure 8.6. val aaaaalvl_s002_t000 (normal): true age 42, recording prediction 42.0 (-0.0 y)

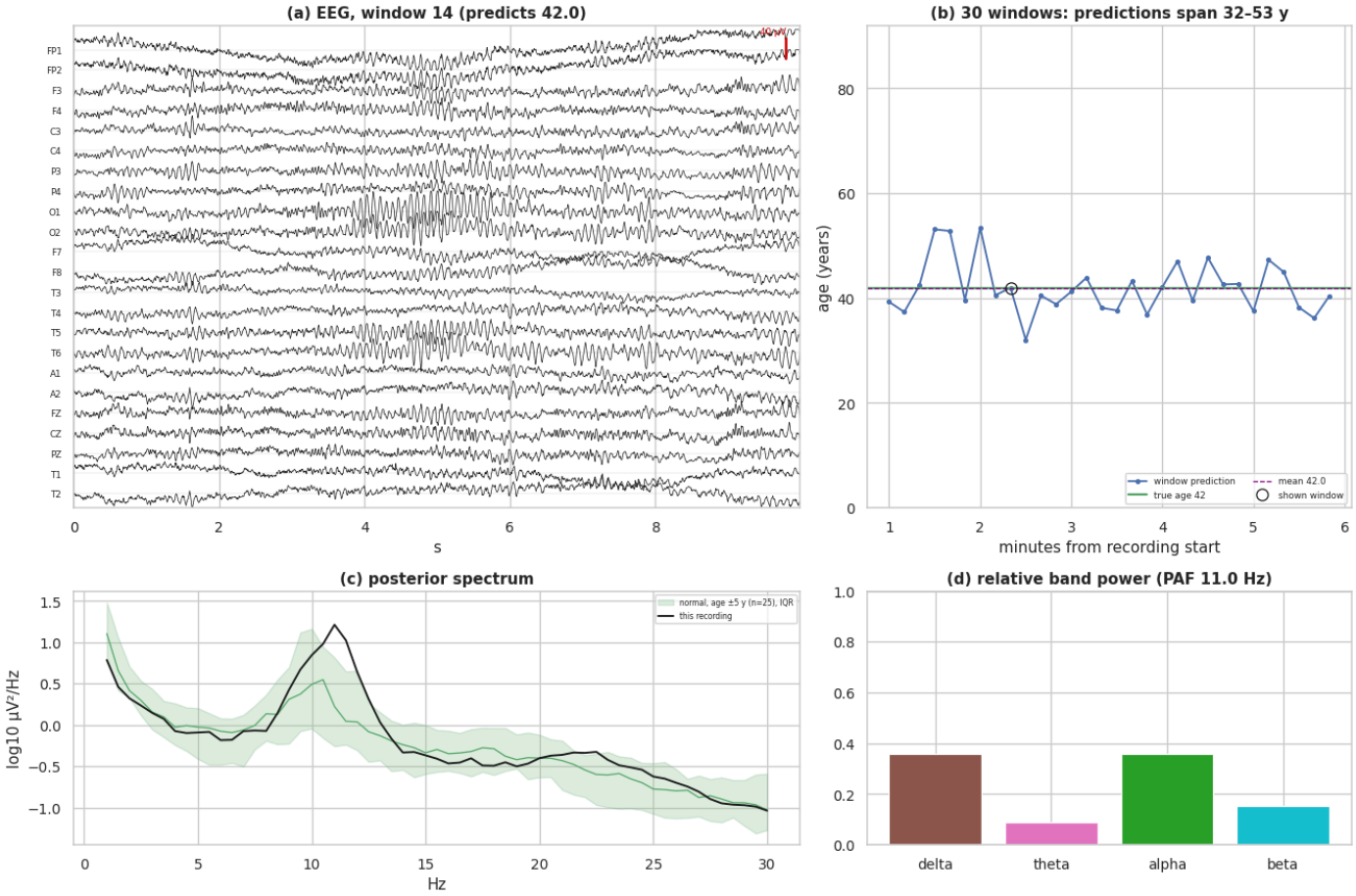


Figure 8.7. val aaaaancc_s001_t000 (abnormal): true age 4, recording prediction 33.6 (+29.6 y)

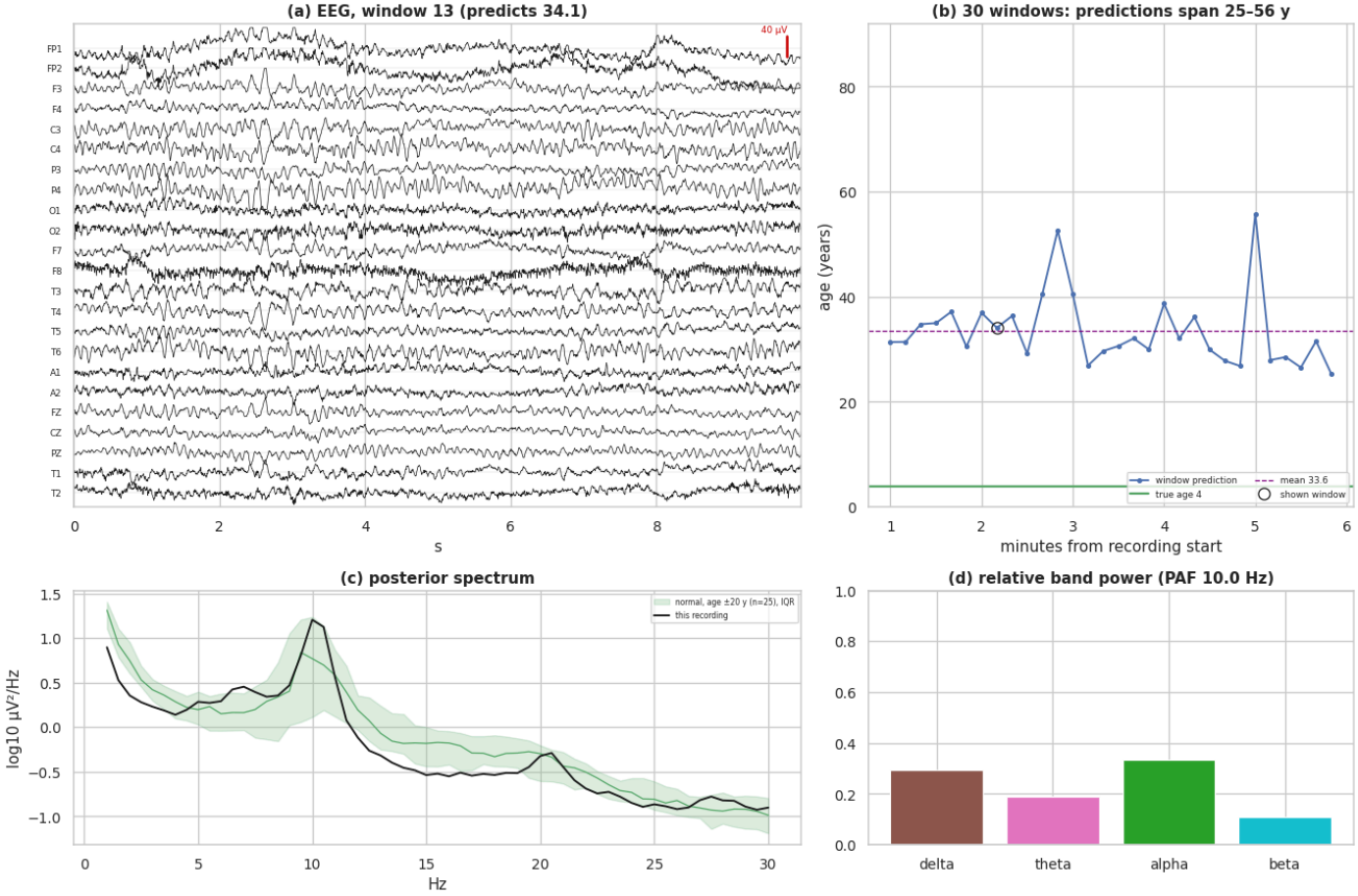


Figure 8.8. val aaaaagxr_s009_t001 (abnormal): true age 27, recording prediction 53.9 (+26.9 y)

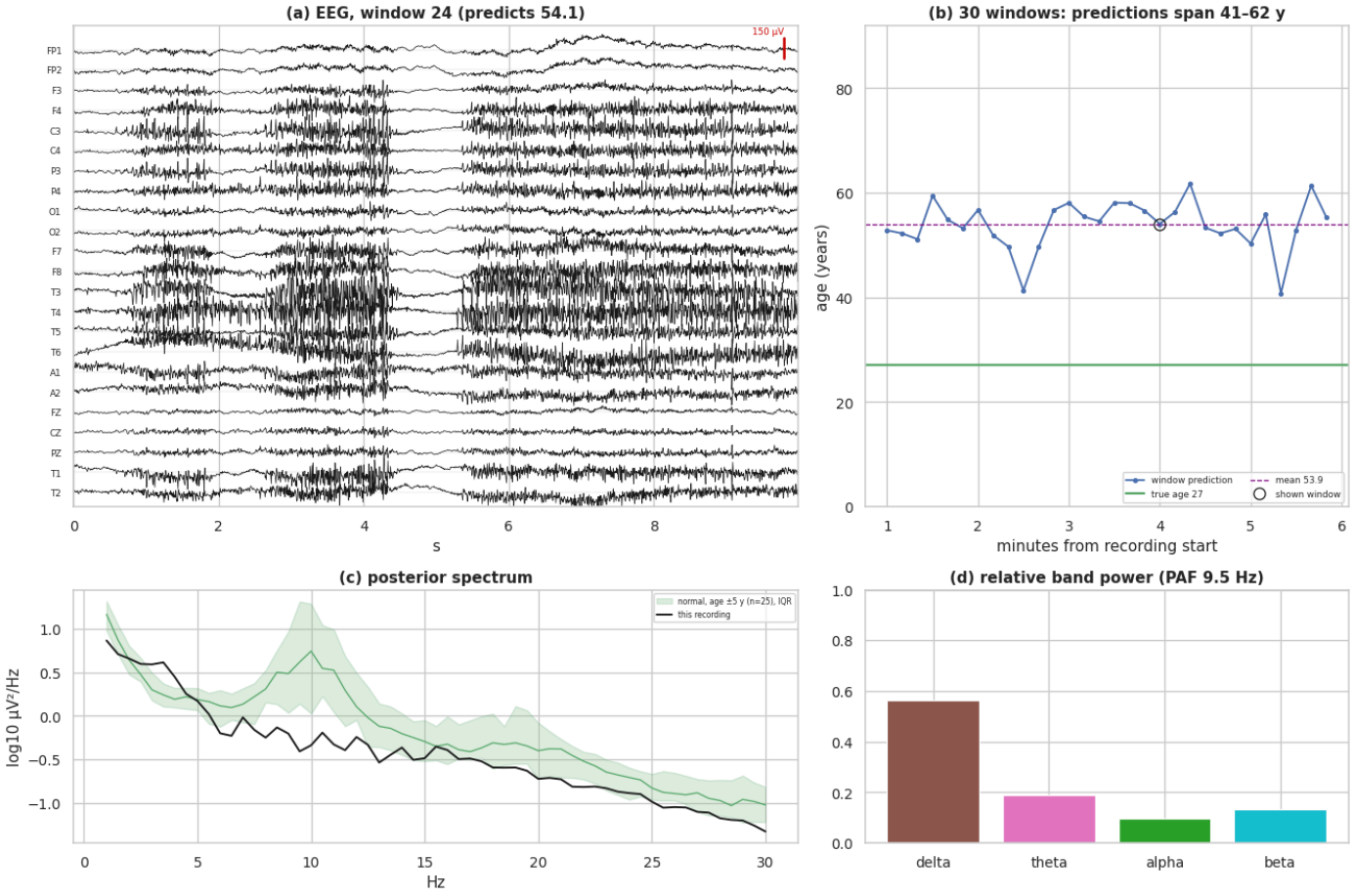


Figure 8.9. val aaaaamtk_s001_t000 (abnormal): true age 79, recording prediction 45.2 (-33.8 y)

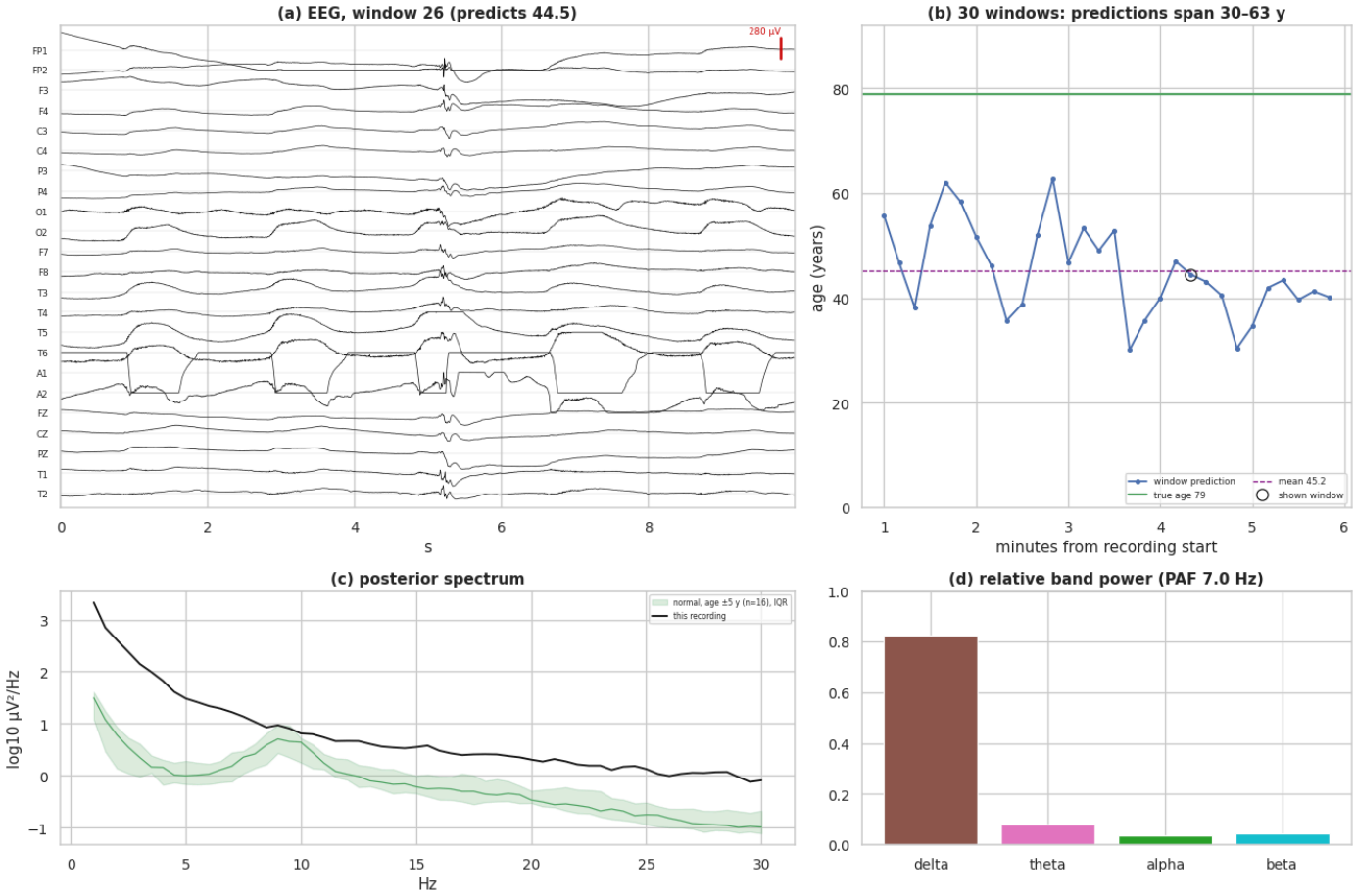


Figure 8.10. val aaaaampk_s001_t000 (abnormal): true age 61, recording prediction 31.0 (-30.0 y)

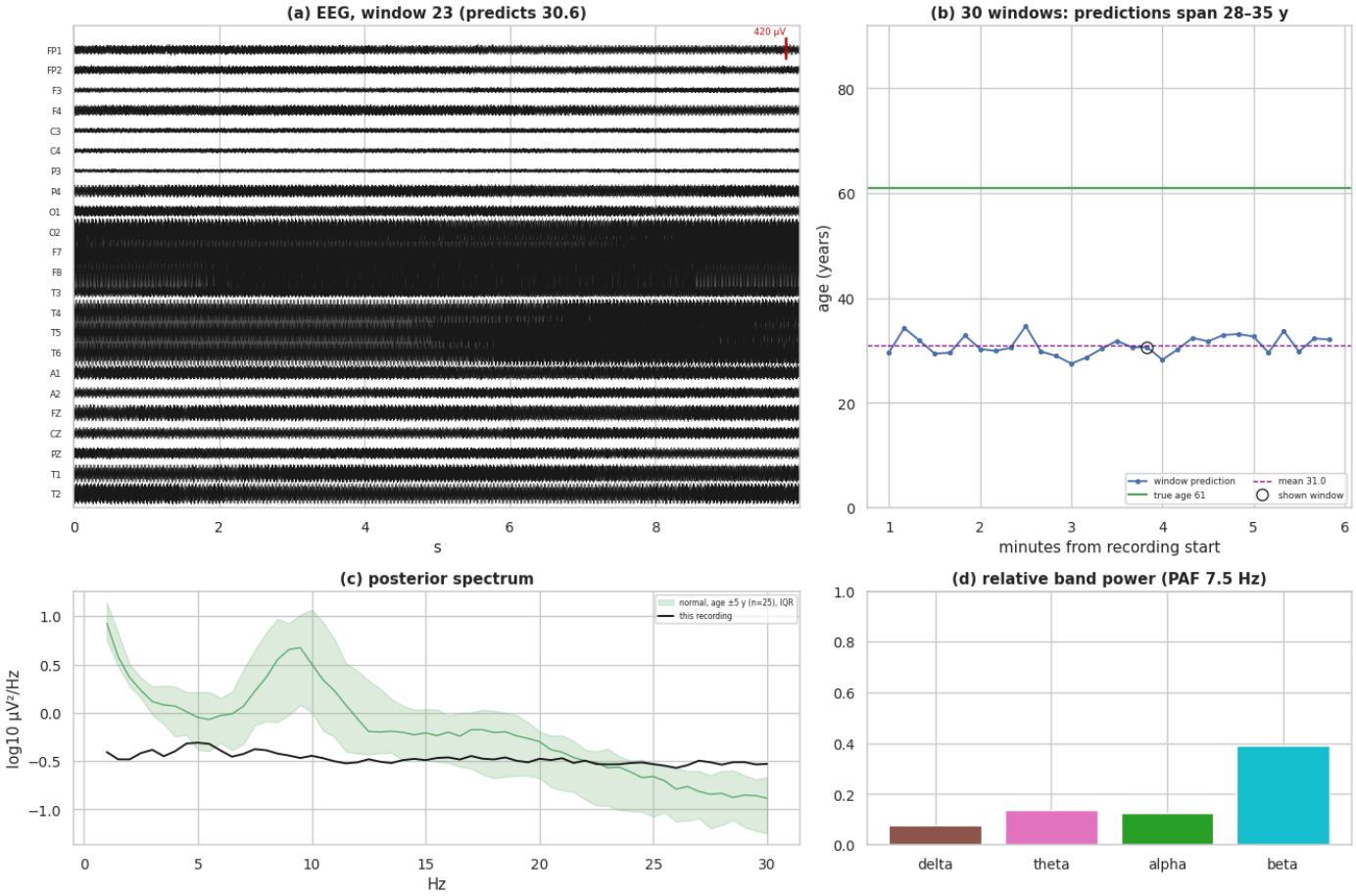


Figure 8.11. test aaaaapof_s001_t000 (normal): true age 37, recording prediction 37.0 (+0.0 y)

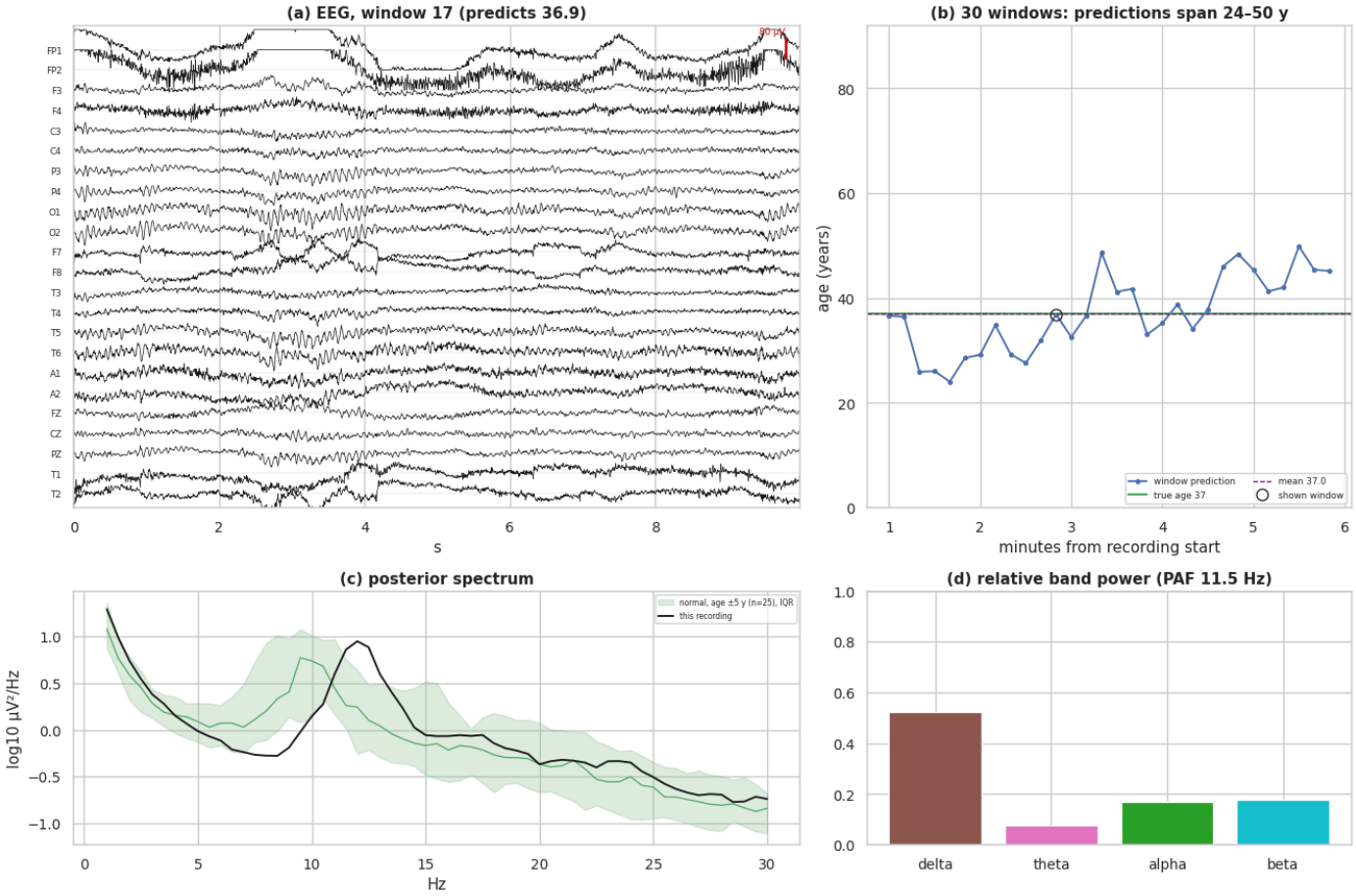


Figure 8.12. test aaaaaoeg_s001_t000 (normal): true age 43, recording prediction 42.9 (-0.1 y)

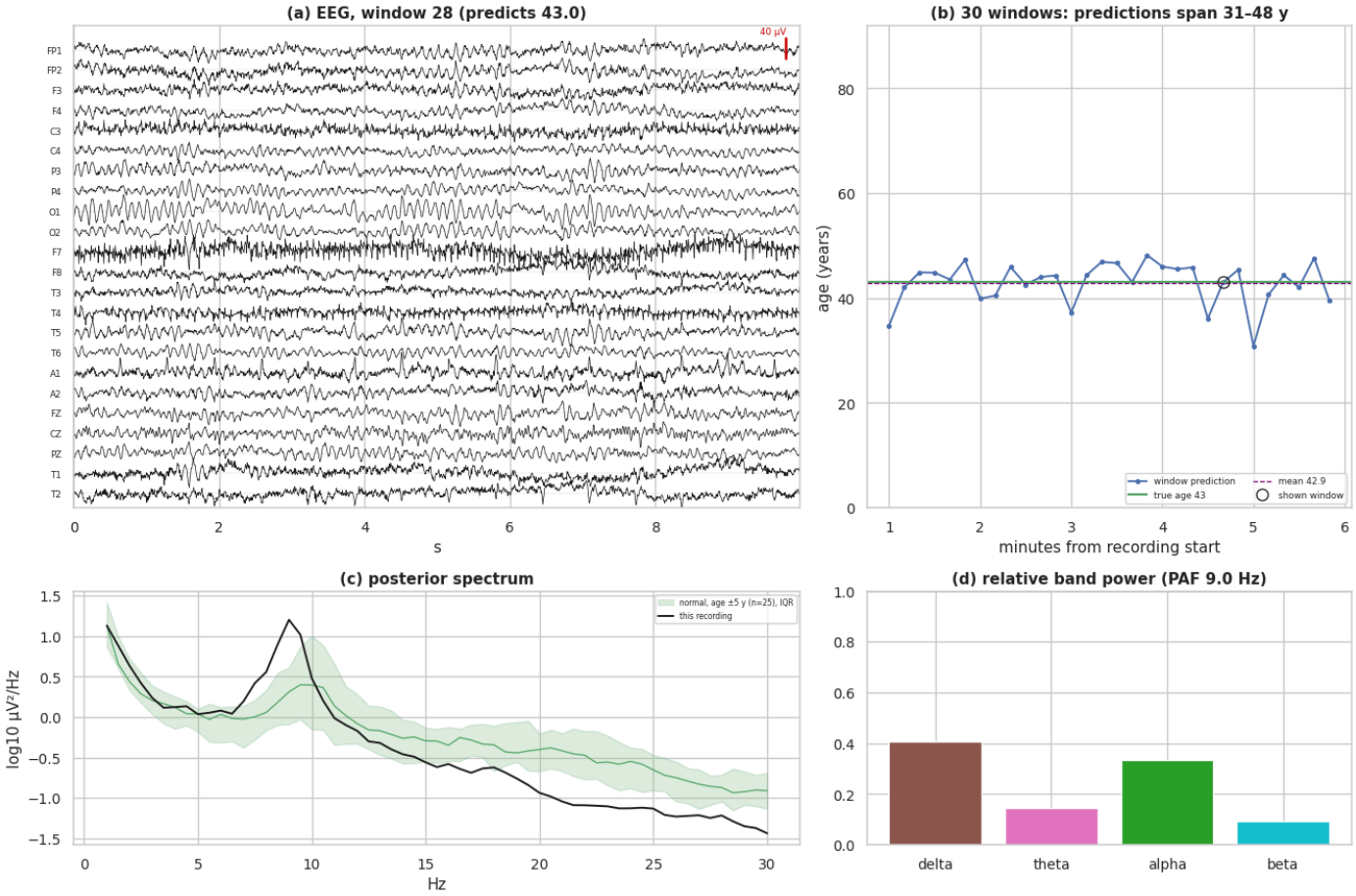


Figure 8.13. test aaaaaohb_s001_t000 (abnormal): true age 11, recording prediction 53.1 (+42.1 y)

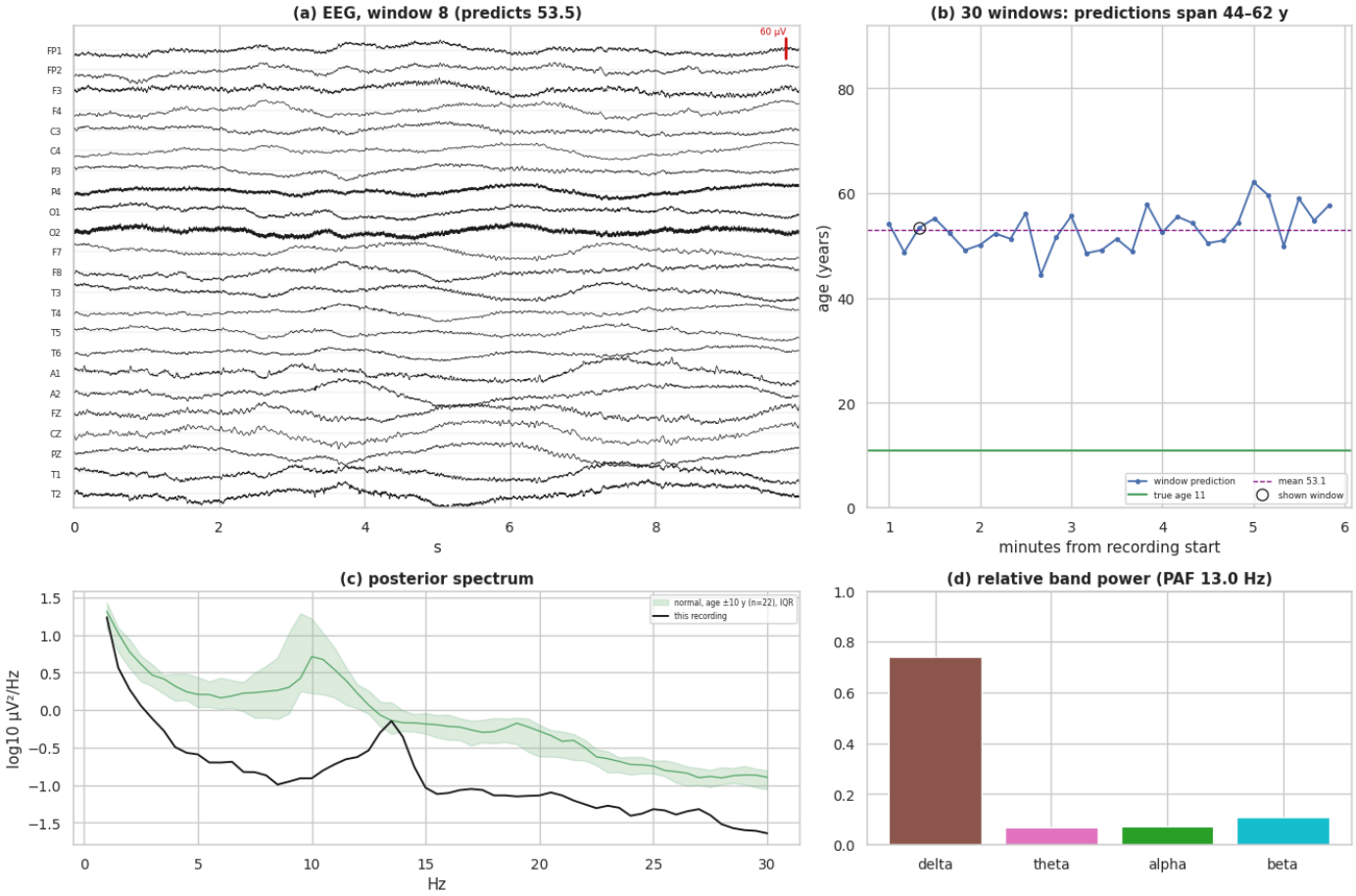


Figure 8.14. test aaaaalvz_s001_t000 (abnormal): true age 22, recording prediction 63.9 (+41.9 y)

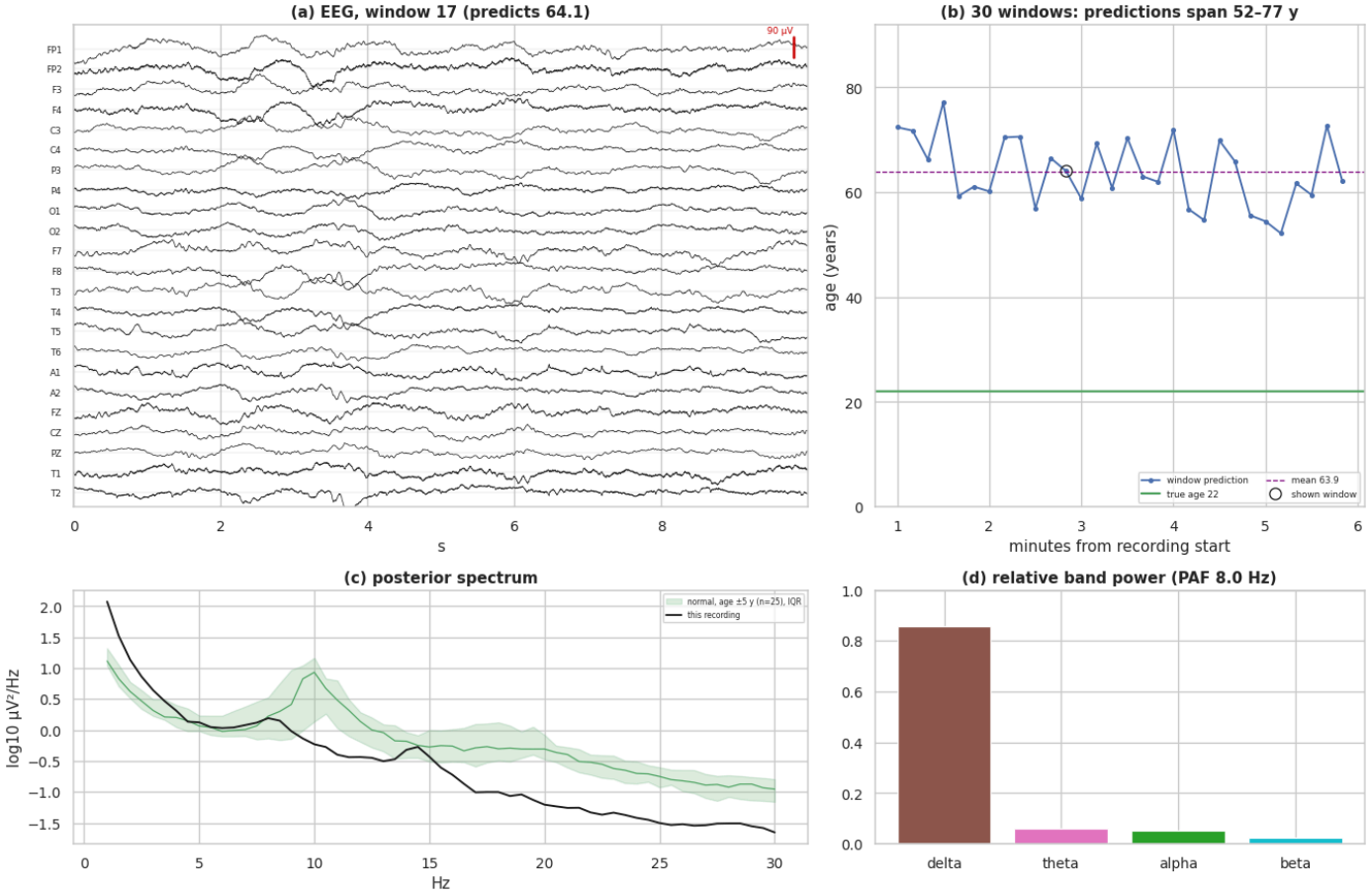


Figure 8.15. test aaaaakxz_s001_t001 (abnormal): true age 78, recording prediction 43.6 (-34.4 y)

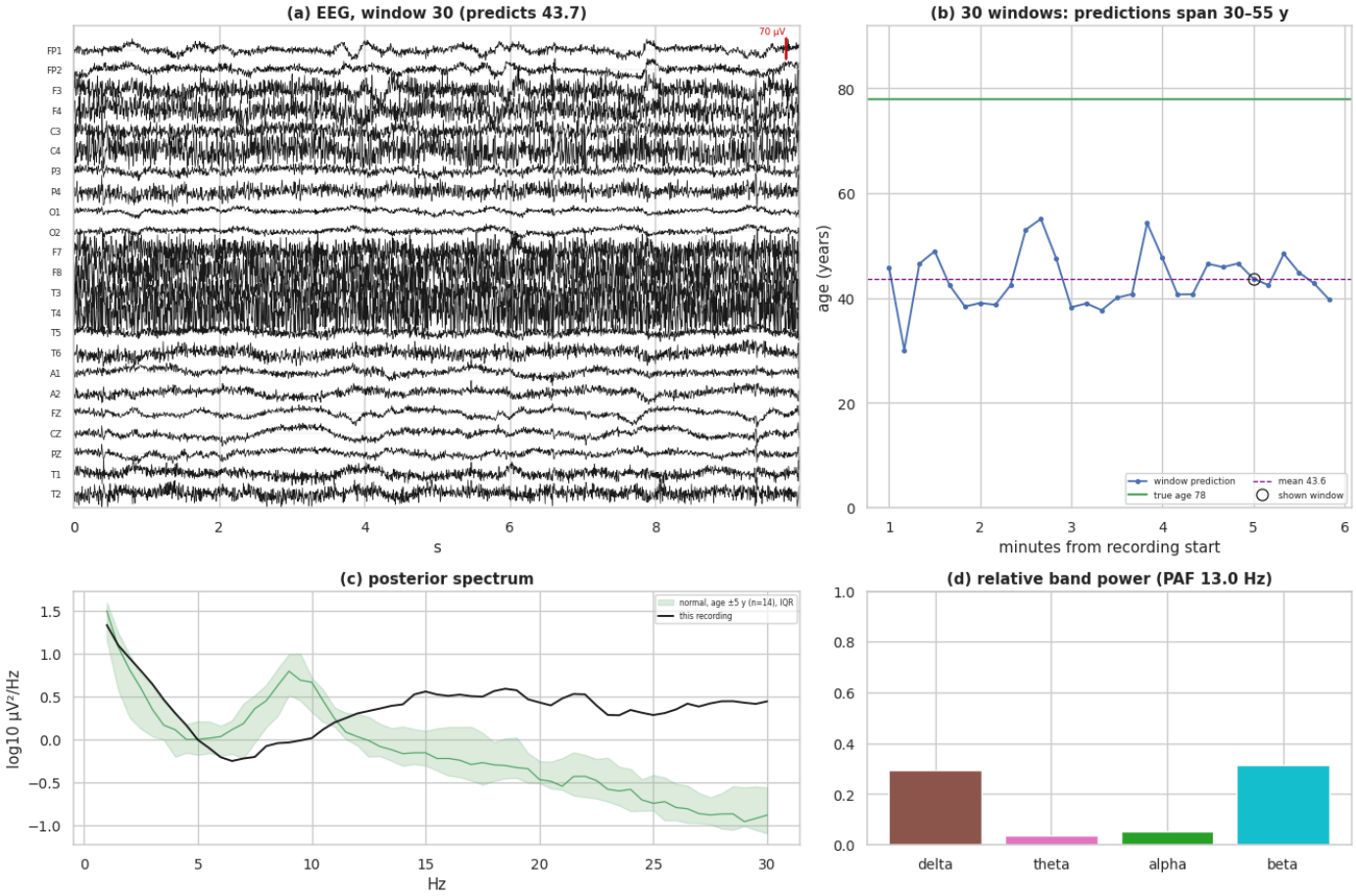
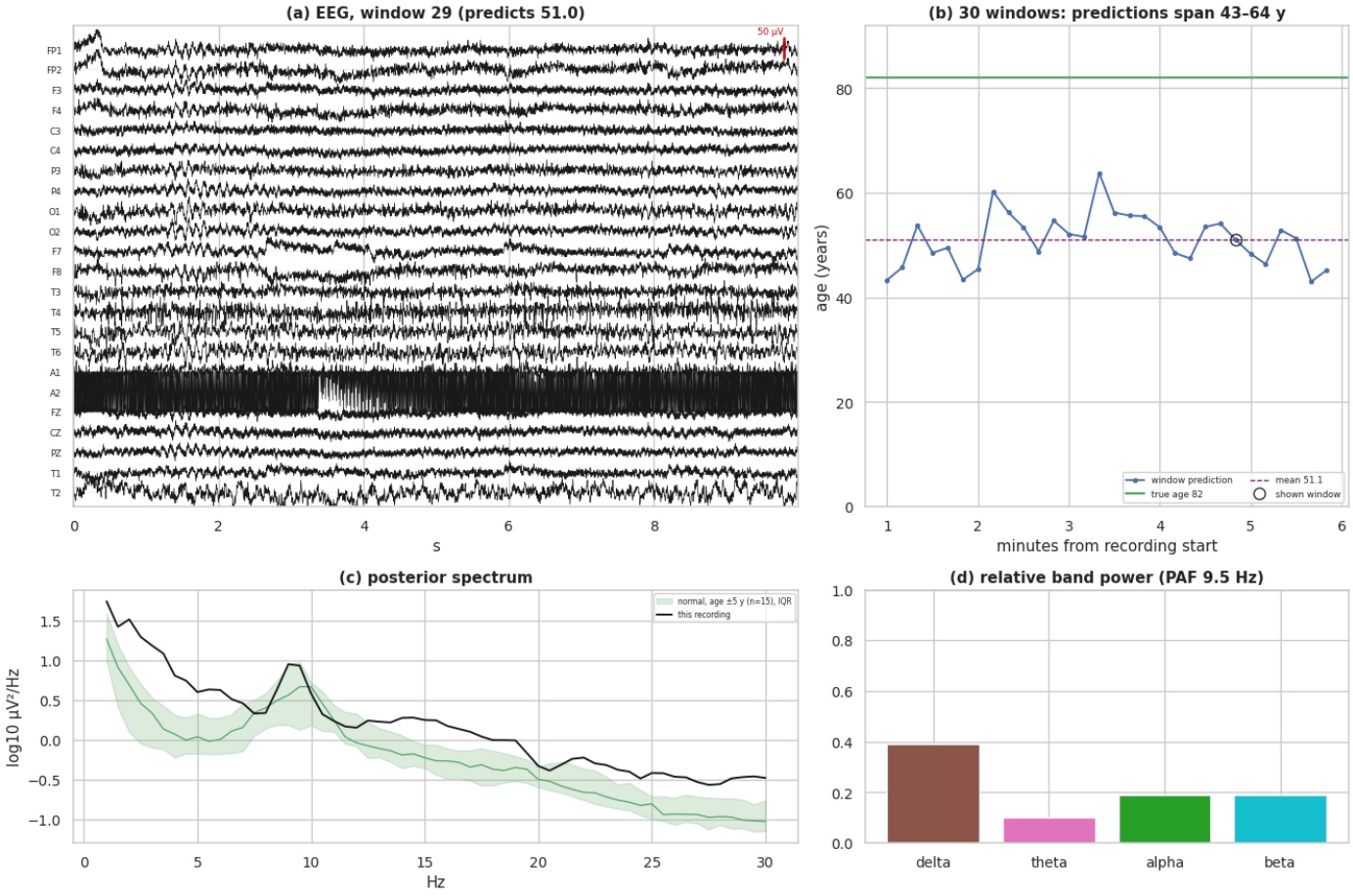


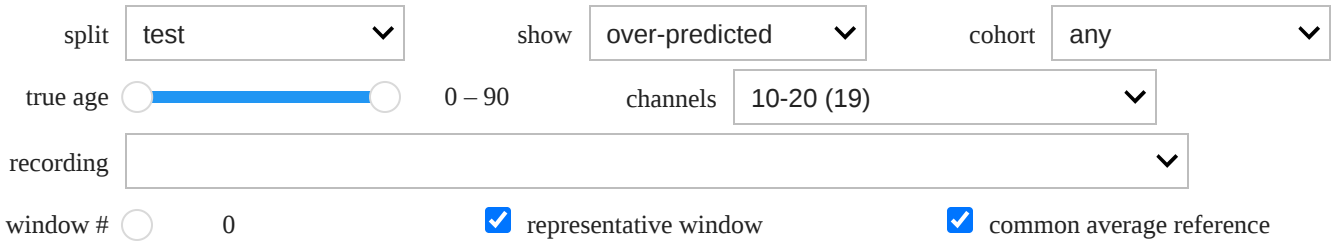
Figure 8.16. test aaaaaiwz_s001_t001 (normal): true age 82, recording prediction 51.1 (-30.9 y)



What the extreme cases show (Table 8.1 and Figures 8.5 onward; M1, as executed). Every one of the most over-predicted recordings is *abnormal* and young: a 4-year-old predicted 34, an 11-year-old predicted 53, and a 22-year-old predicted 64. The 22-year-old's spectrum has about 80% delta power and no alpha peak, which is diffuse slowing that the model reads as an old brain. The most under-predicted are old (61–82 y), mostly abnormal, and are pulled toward the mean. The best-predicted recordings are normal adults in the dense 25–45 range.

8c. Interactive explorer

Pick split, outcome (over/under/accurate), cohort and true-age range; the recording list is sorted by that outcome and shows *true age → predicted (error)*. Untick *representative window* to step through all 30 windows with the slider; *channels* switches between the 10-20 set, posterior only and all 23; *common average reference* and *clip artifacts* toggle the display. Its figure is captioned *Figure 8.E* (explorer). Needs a live kernel; the saved notebook does not keep widget output.



8d. Playground

Edit and re-run. The cells below produce Table 8.2 (candidates), Figure 8.P1 (one recording, posterior channels, first 5 s) and Figure 8.P2 (a three-way comparison at 35). Useful calls:

- `ex.find(age=70, tol=3, kind="under", cohort="normal", split="test", n=10)`: candidates as a table (`kind`: `"accurate"`, `"under"`, `"over"`, or `None` = sorted by age).
- `ex.show(recording, window=None, channels=None, seconds=(0, 10), car=True, clip=1.0)`: one recording. `window` is the position 0–29 in the recording; `channels` is a list such as `["O1", "O2", "PZ"]`; `seconds=(2, 6)` zooms in.
- `ex.compare_at_age(age, tol=3, cohort=None, split=None, channels=STANDARD_19)`: the three-way EEG comparison.
- `ex.windows_of(recording)`: every window's prediction; `ex.signal(split, idx)` returns the raw array (µV, `[23, 2000]`).

Table 8.2. Playground: candidate recordings

Normal recordings near 70 y, most under-predicted first.

	split	recording	cohort	age	pred	err	pred_std	n_windows	paf	rel_delta	rel_alpha
0	test	aaaaajje_s001_t000	normal	72.00	43.21	-28.79	5.76	30	9.00	0.38	0.28
1	test	aaaaajbh_s002_t000	normal	67.00	51.26	-15.74	4.92	30	9.50	0.18	0.32
2	test	aaaaaayx_s002_t000	normal	69.00	55.62	-13.38	3.99	30	9.00	0.42	0.17
3	val	aaaaaoma_s001_t000	normal	67.00	53.97	-13.03	3.81	30	10.00	0.45	0.42
4	test	aaaaalbt_s002_t000	normal	68.00	55.03	-12.97	4.97	30	8.50	0.35	0.22

Figure 8.P1. test aaaaajje_s001_t000 (normal): true age 72, recording prediction 43.2 (-28.8 y)

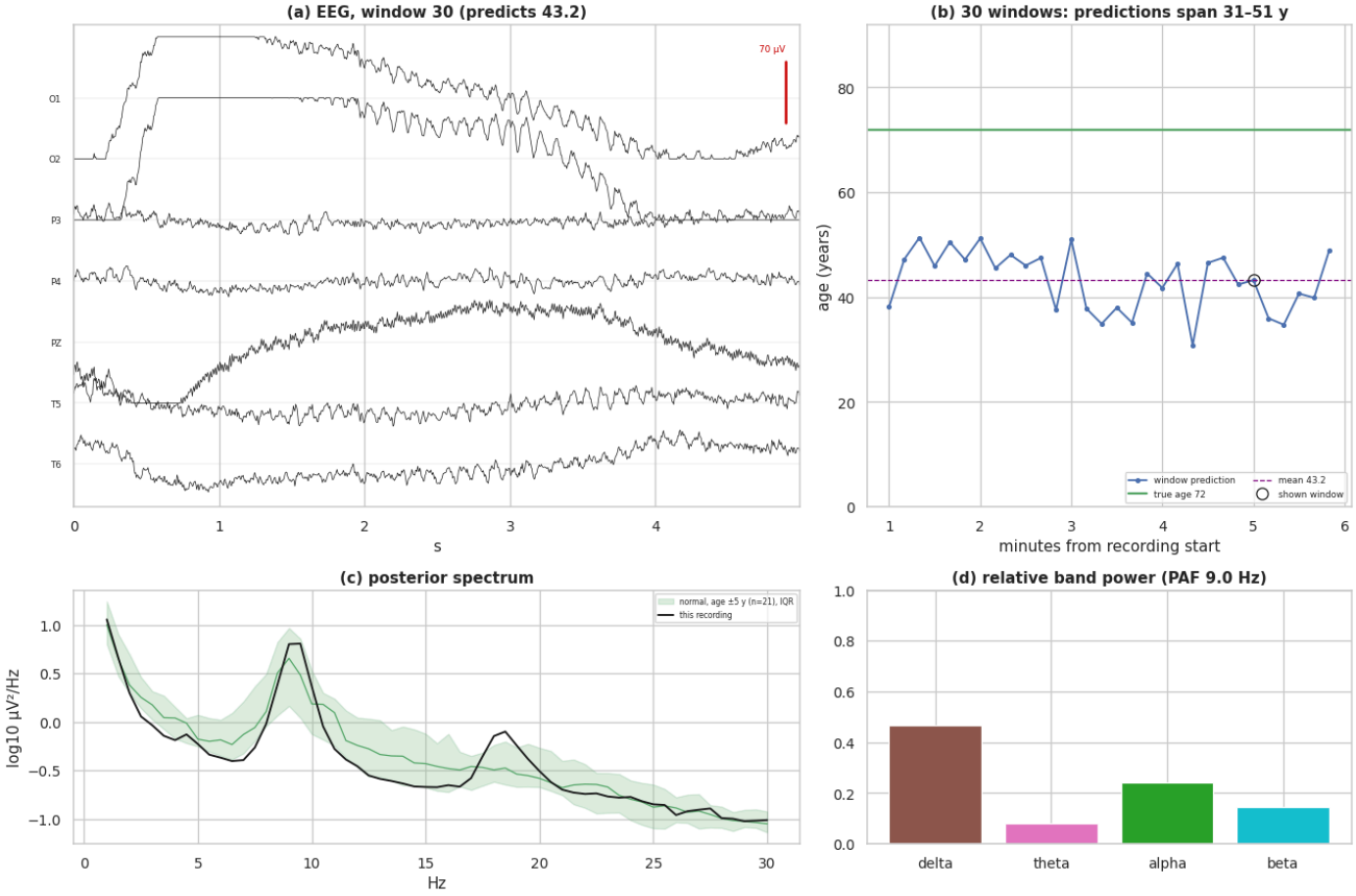
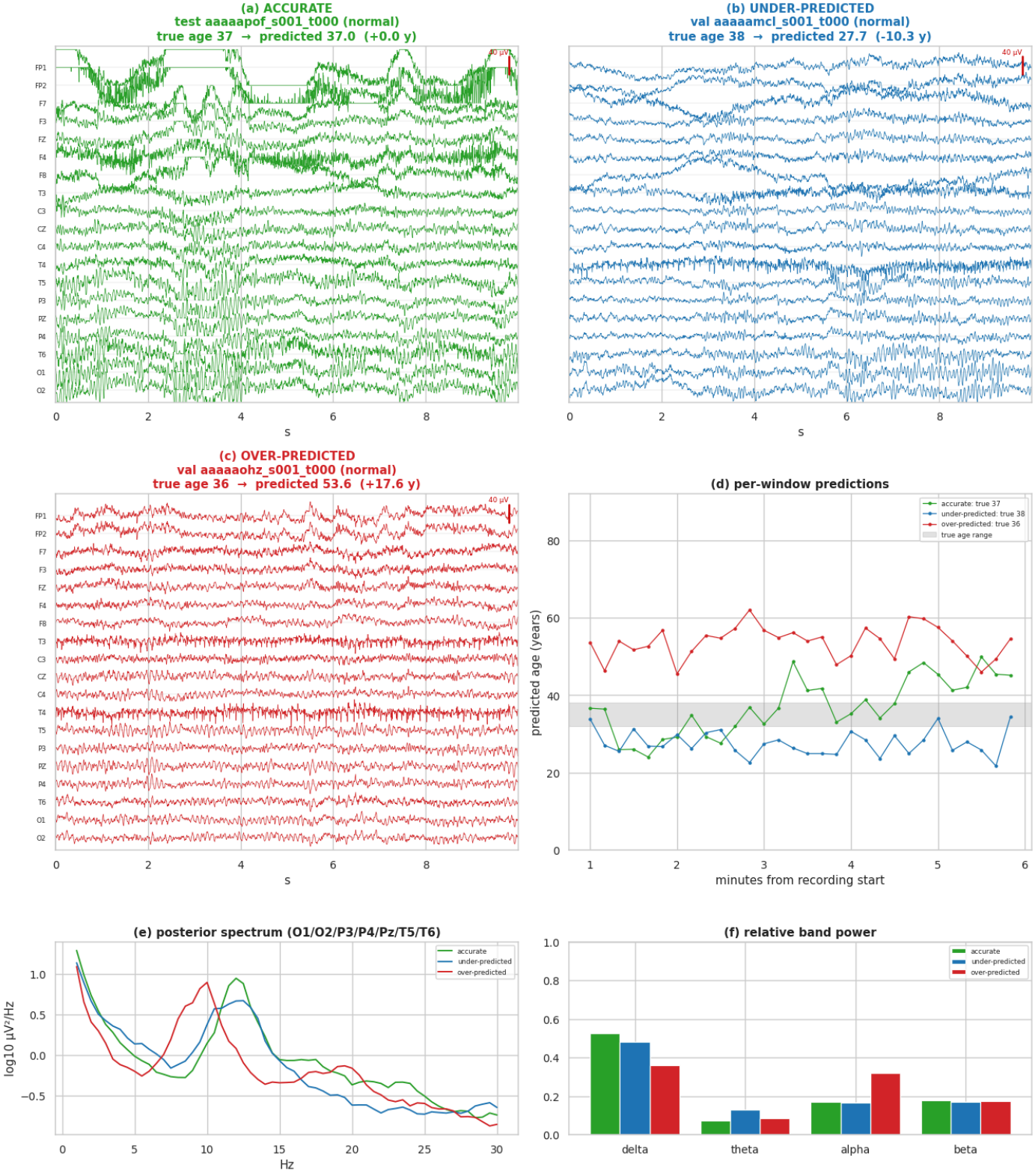


Figure 8.P2. Real EEG at true age ≈ 35: accurate vs. under- vs. over-predicted

CAR, same 40 µV scale in (a)–(c), clipped at ±1 trace spacing; the window closest to each recording's mean prediction.



9. Run comparison: does any variant change the age-bin bias?

The same recording-level analysis for every available local run (val/test predictions are cached per run). The short ablations that ran on SageMaker (EMA, LoRA, soft labels, dropout, …) are listed in Table 9.1 from their ClearML summaries; their checkpoints are not local.

Table 9.1. Short ablations from ClearML

Val/test recording MAE (Eq. (1.5)) and test MAE per cohort (Eq. (1.7)); checkpoints not local.

run	val MAE	test MAE	test normal	test abnormal
E2 baseline (D, 4 ep)	7.68	8.25	–	–
M1 mixup 0.4	7.46	8.45	6.60	10.65
M1c C-Mixup σ=5	7.54	8.23	6.50	10.29
M12 mixup+EMA	7.48	8.45	6.57	10.69
M2 EMA eval	7.58	8.17	6.35	10.35
M3 LoRA r16	7.68	9.24	7.41	11.42
M4 soft labels	7.67	8.59	6.92	10.57
F1 layer decay 0.5	7.74	8.36	6.42	10.68
F2 dropout 0.1	13.25	15.17	13.37	17.32
F3 drop path 0.2	7.70	8.41	6.31	10.91
F4 weight decay 0.1	7.66	8.24	6.49	10.32
G1 head init 1.0	7.76	8.38	6.64	10.45
H1 layer decay 0.75	7.71	8.67	6.64	11.09
H2 layer decay 0.85	7.86	8.54	6.74	10.69

Table 9.2. Local runs, recording level

MAE (2.1), median AE (2.2), R^2 (2.5), bias slope (2.9), and the sample std of the predicted and true ages,

$$\mathrm{sd}(\hat{y}) = \sqrt{\frac{1}{R-1}\sum_{r}\left(\hat{y}_r - \bar{\hat{y}}\right)^2}$$

(a prediction sd below the true sd is regression to the mean).

	mae		median_ae		r2		bias_slope		pred sd		true sd	
split	test	val	test	val	test	val	test	val	test	val	test	val
run												
B1 balance sqrt-inv	8.47	7.83	6.29	6.54	0.61	0.60	-0.37	-0.32	14.60	13.76	18.08	15.93
B2 balance inverse	8.57	7.96	6.60	6.70	0.60	0.58	-0.33	-0.29	15.64	14.62	18.08	15.93
L1 mixup+EMA 12ep	8.47	7.51	6.58	6.22	0.62	0.63	-0.42	-0.37	13.18	12.52	18.08	15.93
M1 mixup	8.45	7.46	6.78	6.14	0.62	0.63	-0.41	-0.37	13.42	12.70	18.08	15.93
M12 mixup+EMA	8.45	7.48	7.06	6.16	0.61	0.63	-0.42	-0.37	13.35	12.62	18.08	15.93
M1c C-Mixup	8.23	7.54	6.59	6.01	0.63	0.62	-0.39	-0.33	13.78	13.41	18.08	15.93

Figure 9.1. Run comparison per true-age decade

Test, per true-age decade (Eq. (5.2)): (a) mean predicted age, (b) MAE.

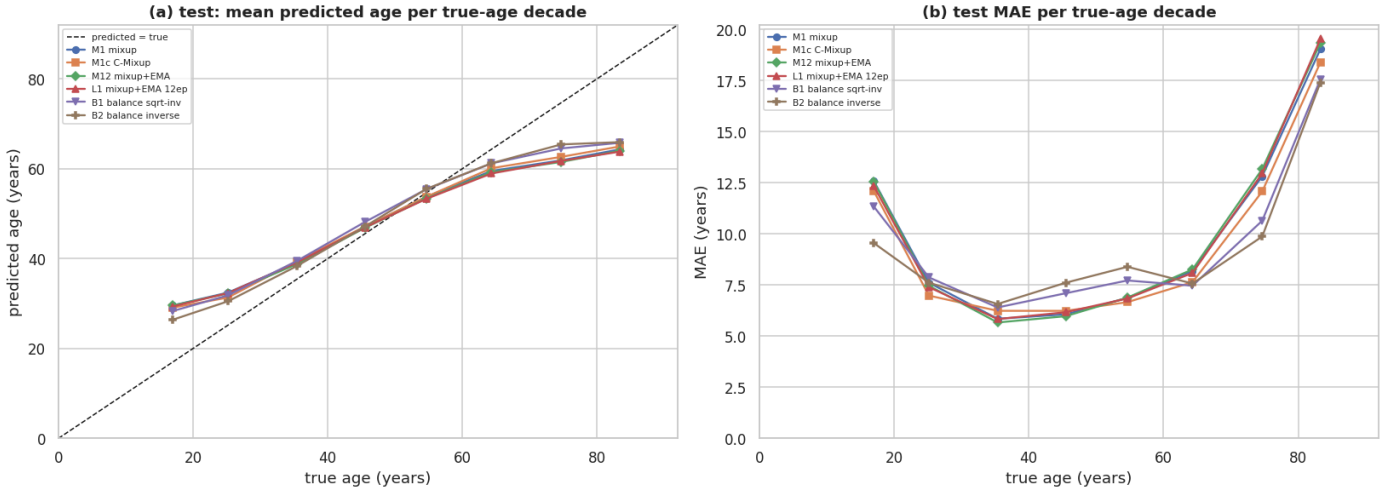


Table 9.3. Mean predicted age per true-age decade

The numbers behind Figure 9.1(a), test recordings.

test decade	M1 mixup	M1c C-Mixup	M12 mixup+EMA	L1 mixup+EMA 12ep	B1 balance sqrt-inv	B2 balance inverse	true (mean age)
10-19	29.6	29.1	29.6	29.3	28.3	26.4	17.0
20-29	32.4	31.3	32.3	32.3	31.8	30.5	25.2
30-39	39.0	39.4	38.7	39.1	39.4	38.3	35.4
40-49	47.1	47.1	46.8	46.9	48.1	47.1	45.5
50-59	53.7	53.7	53.4	53.3	55.4	55.5	54.6
60-69	59.5	60.1	59.2	58.9	61.2	61.2	64.3
70-79	61.8	62.6	61.5	61.7	64.5	65.4	74.7
80-89	64.3	64.9	64.0	63.8	65.7	65.9	83.3

Age-balanced loss (B1 = sqrt-inverse, B2 = inverse weights; short schedule, no mixup)

As executed. The loss of Eq. (0.6) becomes a weighted mean. With N_k the number of train windows in the 1-year age bin $k(i)$ of window i:

$$\mathrm{L} = \frac{1}{B}\sum_{i=1}^{B} \omega_{k(i)}\, \ell_\delta\!\left(\hat{z}_i - z_i\right), \qquad \omega_k \propto \begin{cases} N_k^{-1/2} & \text{B1 (sqrt-inverse)} \\ N_k^{-1} & \text{B2 (inverse)} \end{cases}$$

The weights are capped at `balance_max_weight` × the smallest weight and normalised to mean 1 over the train windows (`losses/regression.py::age_balance_weights`).

The balanced loss moves the tails toward the diagonal, but only partly, and it costs accuracy in the dense middle:

Table 9.4. Balanced loss: test mean predicted age per decade

Test mean predicted age per true-age decade (Eq. (5.2)) for M1 and the two balanced runs.

test decade (true mean age)	M1 mixup	B1 sqrt-inv	B2 inverse
10–19 (17)	predicted 29.6	28.3	**26.4**

test decade (true mean age)	M1 mixup	B1 sqrt-inv	B2 inverse
20–29 (25)	32.4	31.8	30.5
50–59 (55)	53.7	55.4	55.5
70–79 (75)	61.8	64.5	**65.4**
80–89 (83)	64.3	65.7	65.9

- Prediction spread widens (test pred sd 13.4 → 14.6 / 15.6, true 18.1) and the bias slope shrinks (val −0.37 → −0.32 / −0.29).
- Per-decade MAE improves where the weights are large (B2, 10–19: 12.6 → 9.6; 70–79: 12.8 → 9.9) and worsens where they are small (40–59: ~6.0–6.9 → 7.6–8.4). Overall MAE goes up: val 7.83 / 7.96 and test 8.48 / 8.57, against 7.68 / 8.25 for the matching no-mixup baseline E2.
- The 80–89 decade barely moves (+1.5 y). Reweighting cannot create information that the EEG of the oldest recordings does not carry (section 7: slowing and pathology dominate there).

Use it when the downstream use needs less age-dependent bias (e.g. brain-age gap across the age range) more than the lowest MAE. **Not** as a way to lower MAE.

10. TUEG: does training on the whole TUH corpus change the errors?

Everything above uses TUAB, the curated normal/abnormal subset of the TUH EEG corpus: about 3k recordings. **TUEG** is the whole corpus: about 70.8k recordings of 15k patients. TUAB is 4 % of it. The later experiments train on cohort `all_both_v3`: TUAB plus every TUEG recording of patients outside the TUAB test set. That is 25,613 training recordings of about 12k patients, ages 18–89, with normal, abnormal and unlabelled recordings (`notebooks/brain_age_research_summary.ipynb`, Section 1).

Two of those models are compared here with M1, on **the same 274 TUAB test recordings**:

- **cvbase:** EEG only. M1's recipe on `all_both_v3`, with the speed settings of the TUH runs.
- **cvctrl:** the best overall model. Like cvbase, but the head reads the class token and also receives the recording's status and pathology vector, so it is not EEG-only.

Both are 5-fold cross-validated, patient-grouped. Their test prediction is the **median of the 5 fold models**. Their per-recording predictions come from `scripts/eval_cv_folds.py` (`$LABRAM_CV_EVAL/cv_preds.json`, outside git). The metrics are those of Section 2, on recordings: MAE (Eq. (2.1)), mean error (Eq. (2.7)), R^2 (Eq. (2.5)) and bias slope (Eq. (2.9)).

Table 10.1. Training recordings per true-age decade: TUAB versus TUAB + TUEG

Training recordings per decade of true age: TUAB's train split (M1) against `all_both_v3`'s (cvbase, cvctrl), and their ratio. Needs the TUH data volume.

true age	TUAB train (M1)	TUAB + TUEG train (cvbase, cvctrl)	ratio
<20	57	282	4.9
20–29	315	3206	10.2
30–39	287	2960	10.3
40–49	405	4122	10.2
50–59	448	5671	12.7
60–69	333	4809	14.4
70–79	190	2932	15.4
80+	110	1631	14.8
youngest age	1	18	–

Table 10.2. TUAB test: TUAB-only M1 versus the TUAB + TUEG models

The 274 TUAB test recordings, all and per cohort (149 normal / 125 abnormal):

- `MAE` (Eq. (2.1));
- `bias` = mean error (Eq. (2.7)); negative means predicted too young;
- `R²` (Eq. (2.5));
- `β` = bias slope (Eq. (2.9)).

M1 is one model. cvbase and cvctrl are 5-model median ensembles.

model	cohort	n	MAE	bias	R²	β
M1 (TUAB only)	**all**	274	8.45	-1.40	0.62	-0.41
	normal	149	6.60	-0.60	0.73	-0.35
	abnormal	125	10.65	-2.35	0.39	-0.56
cvbase (TUAB + TUEG, EEG only)	**all**	274	7.44	-1.21	0.70	-0.37
	normal	149	5.75	-0.56	0.80	-0.30
	abnormal	125	9.46	-1.99	0.52	-0.50
cvctrl (TUAB + TUEG, + status/pathology)	**all**	274	7.10	-0.77	0.72	-0.35
	normal	149	5.31	-0.58	0.82	-0.29
	abnormal	125	9.23	-1.00	0.53	-0.50

Figure 10.1. TUAB test per true-age decade: M1 versus the TUAB + TUEG models

The 274 TUAB test recordings, grouped by true-age decade (as in Eq. (5.2)):

- (a) The mean predicted age against the decade's mean true age. The dashed diagonal is a perfect prediction; a curve flatter than it is regression to the mean.
- (b) The MAE per decade (Eq. (5.2)).

The `<20` decade holds only 6 test recordings.

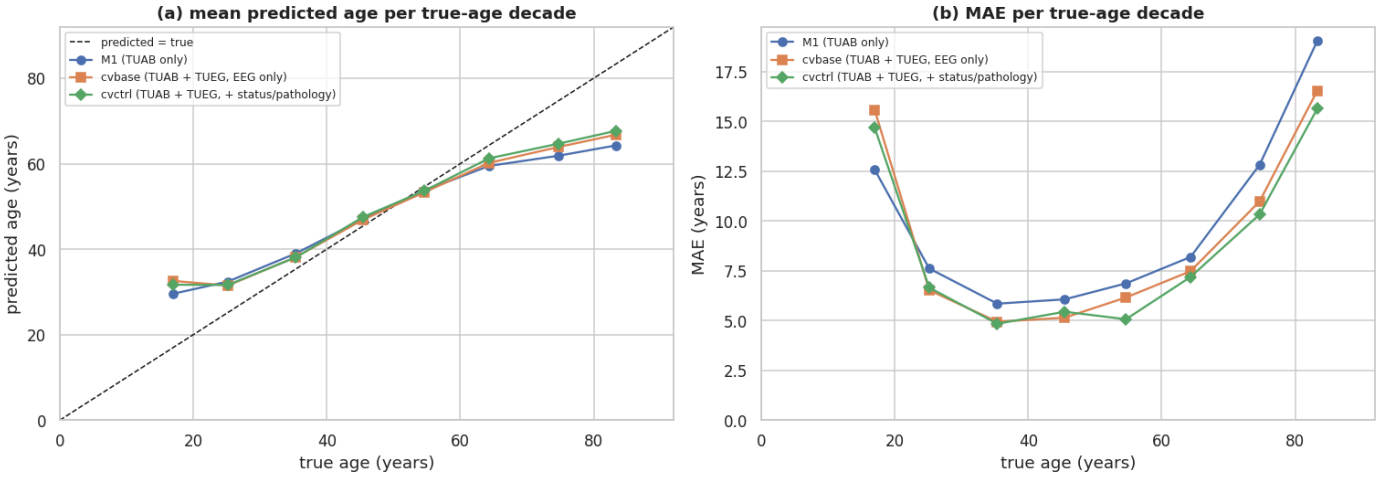
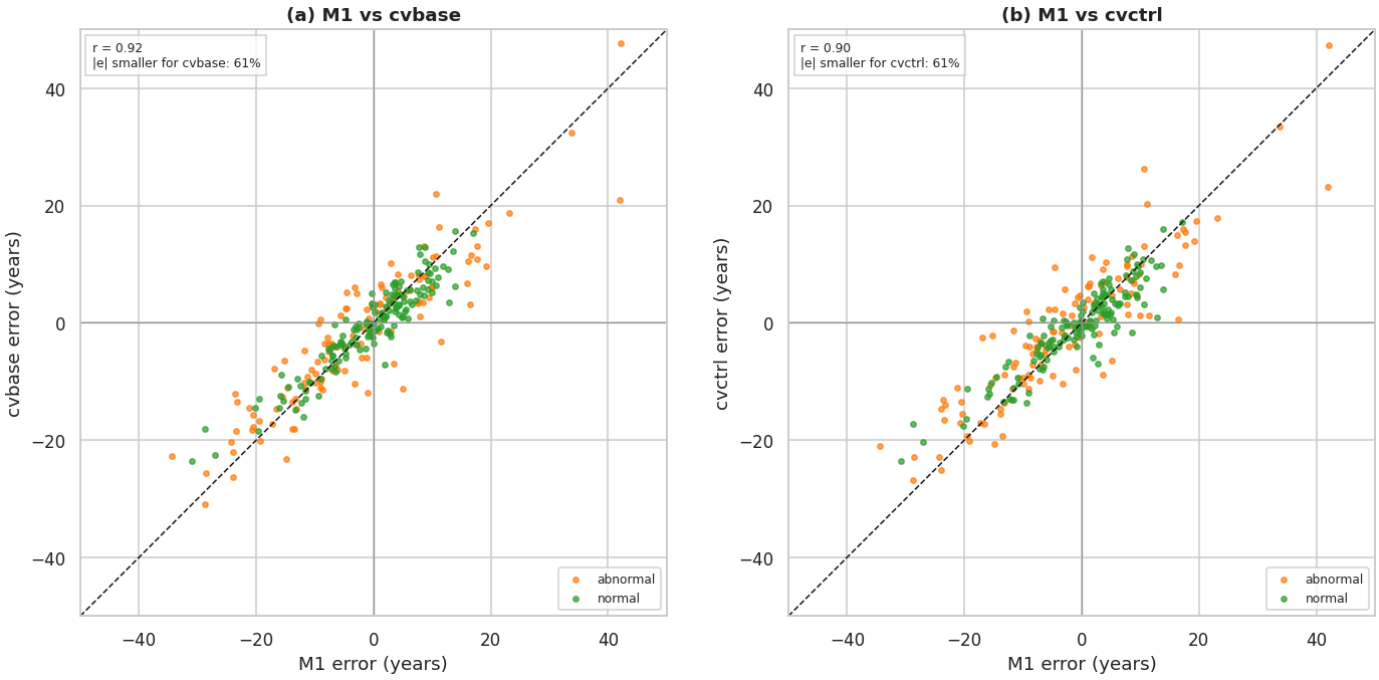


Figure 10.2. Do the TUAB-only and the TUAB + TUEG models fail on the same recordings?

Per TUAB test recording, M1's error $e = \hat{y} - y$ (years) against the error of the two TUAB + TUEG ensembles, coloured by cohort:

- (a) cvbase;
- (b) cvctrl.

Points on the dashed diagonal share the same error. The box gives the Pearson correlation of the errors (Eq. (2.6) applied to e).



What TUEG changes (Tables 10.1, 10.2; Figures 10.1, 10.2; as executed).

- **More data, everywhere but the young.** TUAB + TUEG has 10–15× more training recordings per decade from 20 to 89 (Table 10.1). The relative gain is largest at 60+, the

ages TUAB has least of. The cohort keeps only ages 18–89, so the children of TUAB's train split are gone.

- **Lower error.** On the same 274 TUAB test recordings, the MAE drops from 8.45 (M1) to 7.44 (cvbase, EEG only) and 7.10 (cvctrl). On normal recordings it drops from 6.60 to 5.75 / 5.31, below the best published TUAB normal result (6.60, Gemein et al. 2024).
- **Better in every decade from 20 to 89** (Figure 10.1(b)), by 0.7–1.8 years in the dense middle (30–59) and by up to 3.4 years at 80+. **Worse under 20**, because the TUEG models never trained on children.
- **The error pattern of Sections 5–7 survives.** Regression to the mean weakens only a little (β −0.41 → −0.37 / −0.35). The old recordings are still predicted much too young, despite about 15× more training examples: 10–11 years too young at 70–79 and 16–17 years at 80+. Abnormal recordings remain the hard cohort (MAE 9.2–9.5). This supports the conclusion of Section 7: the remaining error is not mainly a lack of old training examples. It is what the EEG carries — pathological slowing reads as age.
- **Same recordings, smaller errors** (Figure 10.2). M1's and the TUEG models' errors correlate at r = 0.90–0.92, so the recordings that M1 gets wrong are largely the ones the bigger models get wrong. The TUEG models have the smaller error on 61 % of the recordings. The TUEG models shrink most errors rather than fixing a different set of recordings.

11. Takeaways

The narrative is written against the executed outputs above; equation, table and figure numbers refer to them. Rerun the notebook after new runs finish; section 9 picks them up automatically.

```
* Memorization: recording MAE train 4.31 (eval mode) vs val 7.46 / test 8.45.
* Mean vs median: test MAE 8.45, median AE 6.78, P90 17.72; the tail is concentrated in
the oldest bins and abnormal recordings.
* Regression to the mean: test bias slope -0.41; the 10-19 decade (mean age 17) is pred
icted 30, the 80-89 decade (mean age 83) is predicted 64; predictions span sd 13.4 vs t
rue sd 18.1.
* Calibration: on val, age ≈ 1.00·pred +0.46, so predictions are already E[age | pred].
Cole's inversion removes the slope (-0.08) but raises test MAE to 9.94.
* Cohort: test MAE normal 6.60 vs abnormal 10.65.
```

Interpretation

- **The bias is not a calibration bug** (Table 6.1). Linearly regressing age on the prediction (Eq. (6.1)) gives slope $\alpha \approx 1$, so the model already outputs roughly E[age | EEG]. Regression to the mean is what a squared/Huber-loss estimator does when the EEG carries limited age information. No deployable post-hoc correction lowers the MAE. Cole's correction trades a flat bias for a higher MAE, and the "age-level" correction only looks good because it uses the true age.

- **The errors concentrate in the tails of the age distribution and in the abnormal cohort** (Figures 3.1, 5.1). The 10% worst windows carry about 45% of the Huber loss. Sampling density explains part of it: the bins with the fewest train windows (<20 y: ~8k; 70–79: 24k; 80–89: 13k, vs. ~40–55k for 20–69) have the largest MAE. But density is not the whole story. 20–29 and 60–69 have nearly the same train counts (~40k windows), yet on test the 20–29-year-olds are predicted ~7 years too old and the 60–69-year-olds ~5 years too young, and 80–89 is worse than 10–19 despite more data. Even on **train** the predicted-vs-true fit slope is 0.80, so the shrinkage is built into the fit, not only a val/test effect.
- **Pathological slowing reads as "older"** (Table 7.2, Figure 7.1). After removing the age trend, the residual correlates with lower posterior PAF ($\rho \approx -0.2$) and a higher theta/alpha ratio (+0.17). Abnormal recordings sit at a median PAF of 7.5 Hz vs. 10 Hz for normal ones. Among age-matched 40–69-year-olds, 72% of the most over-predicted fifth are abnormal vs. 43% of the most under-predicted. Within the abnormal cohort, delta power and theta/alpha drive the over-prediction; within the normal cohort the effect is weak. The model partly confuses disease-related slowing with ageing. This is a property of the data rather than a modelling bug, so the brain-age gap on abnormal EEG is partly a pathology signal.
- **Within-recording spread is a weak confidence signal** ($r \approx 0.1$ with the error), and median pooling does not beat mean pooling.
- **Levers that can actually move the tails**: an age-balanced loss (B1/B2, section 9), which accepts a higher MAE in the dense middle for less tail bias; longer training with mixup+EMA (L1); and, for the abnormal cohort, conditioning on or modelling pathology (e.g. a normal/abnormal auxiliary head) instead of hoping the age head ignores it.